

Leverage Regulation and Market Structure: A Structural Model of the U.K. Mortgage Market

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ABSTRACT

I develop a structural model of mortgage demand and lender competition to study how leverage regulation affects the U.K. mortgage market. Using variation in risk-weighted capital requirements across lenders and mortgages with different loan-to-values (LTVs), I show that a 1-percentage-point increase in risk-weighted capital requirements increases lenders' marginal cost of originating mortgages by about 26 basis points (11%) on average. I use the estimated model to study proposed leverage regulations. Counterfactual analyses show that large lenders exploit a regulatory cost advantage, which increases concentration by about 20%, and suggest that banning high-LTV mortgages may reduce large lenders' equity buffer.

AS A RESULT OF THE GLOBAL financial crisis, several European countries and the United States introduced new regulations to prevent excessive bank leverage and increase the stability of the financial system. Prominent examples

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include risk-weighted minimum capital requirements for lenders and limits to loan-to-income (LTI) and loan-to-value (LTV) for borrowers (Acharya, Engle, and Pierret (2014), Behn, Haselmann, and Wachtel (2016a), Jiménez et al. (2017), DeFusco, Johnson, and Mondragon (2020)). While there is relatively wide agreement on the necessity of such measures, the design of lender- and borrower-based leverage regulations and their implications for the allocation of credit have been subject to debate.

To better link capital to risk, policymakers have allowed lenders to invest in internal rating-based (IRB) models to calculate the risk weights on their activities. IRB models may improve risk management practices by financial intermediaries, but they are costly to develop and maintain. As a result, a two-tier system has emerged whereby large lenders adopt IRB models, while small lenders use the standardized regulatory approach. This heterogeneity across lenders and asset classes can have unintended consequences, such as potential regulatory arbitrage and reduced competition (Acharya, Schnabl, and Suarez (2013), Behn, Haselmann, and Vig (2016b), Greenwood et al. (2017), Buchak et al. (2018b)).¹

In this paper, I develop a structural model to quantify the cost of risk-weighted capital requirements and the equilibrium impact of heterogeneous leverage regulations on interest rates, concentration, and risk in the mortgage market, which played a central role in the global financial crisis (Mian and Sufi (2011), Campbell and Cocco (2015), Corbae and Quintin (2015)). To capture the richness in product differentiation, households' choices, and lenders' capital requirements in modern mortgage markets, I take an approach inspired by the industrial organization literature on differentiated product demand. Specifically, I estimate my model using loan-level data on the universe of mortgage originations in the United Kingdom and a novel identification strategy that exploits variation in leverage regulation both across lenders and across mortgages with different LTVs.²

I begin by providing descriptive evidence that large lenders adopting IRB models have lower risk weights than small lenders across almost all mortgages they originate. I also show that the difference in risk weights is larger for mortgages with lower leverage and cannot be explained by observable ex-ante borrower characteristics or ex-post mortgage performance. As a result, large lenders have an absolute advantage across all mortgages they originate and a relative advantage across mortgages with lower leverage, all else equal. I label this phenomenon the regulatory-risk-weight gap.

¹ The fact that different lenders can calculate risk-weighted capital requirements using different approaches has been blamed as one of the main barriers to competition in some key U.K. banking markets, such as mortgages and personal current accounts (Competition and Markets Authority (2015), Financial Conduct Authority (2018)).

² The United Kingdom has been at the forefront of policies to regulate both lenders' and households' leverage. Since the 1990s, U.K. regulators have imposed time-varying, bank-specific minimum capital requirements on banks (Aiyar et al. (2014)). More recently, borrower-based macroprudential policies such as LTI limits have been introduced (Bank of England (2014)).

I next explore how heterogeneity in risk-weight regulation affects banks' ability to fund themselves cheaply, which is their most important competitive edge (Hanson, Kashyap, and Stein (2011)). First, I show that the large differences in regulatory risk weights affect lenders' pricing behavior and their specialization patterns in different LTV segments of the mortgage market. Second, I exploit variation from the approval of an IRB model for a medium-size lender in an event-study setting to study the effect of changes in risk weights on mortgage prices.³

Overall, my descriptive evidence suggests that leverage regulations affect lenders' pricing and mortgage originations, consistent with recent work in other countries and markets (Behn, Haselmann, and Vig (2016b), Gropp et al. (2019)). However, understanding the equilibrium impact of leverage regulations using a purely reduced-form approach faces the challenge of isolating *observed* changes in regulations from contemporaneous *unobserved* changes in market power and business strategies, which can interact with regulatory changes and confound their effects. Furthermore, regulatory changes are endogenous, and different regulations are often adjusted simultaneously, making identification more complex (Kovner and Van Tassel (2018)).

Motivated by the descriptive evidence, I develop a workhorse structural model that allows me to recover borrowers' demand elasticities and lenders' unobservable costs of originating mortgages. On the demand side, I assume that borrowers have heterogeneous preferences over interest rates and other characteristics (rate type, lender, and maximum leverage) and choose the mortgage product that gives them the highest utility across their individual-specific menu. I depart from traditional individual-level discrete-only choice models and allow borrowers to also choose the loan amount, following Dubin and McFadden (1984). The discrete-continuous choice allows me to decompose the elasticity of demand to the interest rate into a product elasticity and a loan demand elasticity, both affecting substitution patterns and lenders' markups.

On the supply side, I model lenders as heterogeneous multiproduct firms that offer differentiated mortgages and compete on interest rates, subject to regulatory leverage constraints. I use the elasticity parameters from the demand estimation together with lenders' Lerner formula for profit-maximizing prices and loan-level data on arrears to back out unobservable marginal costs of originating mortgage products.

I estimate the model using the universe of mortgage originations in the United Kingdom over the period 2015 to 2016. The United Kingdom represents an ideal laboratory to estimate my model given the rich variation in mortgage products and risk-weighted capital requirements. I estimate the demand-side parameters using variation in product characteristics and borrower choices. I address endogeneity problems arising from the simultaneous nature of the

³ In the [Internet Appendix](#), which is available in the online version of the article on *The Journal of Finance* website, I provide further evidence on the relation between risk weights and mortgage rates. I also exploit the June 2014 introduction of a maximum LTI limit in the United Kingdom to study the effect of leverage regulation on mortgage originations.

discrete-continuous choice and from omitted variables by instrumenting endogenous rates using predetermined risk-weighted capital requirements as cost shifters.

I estimate the supply-side parameters using variation in product characteristics that affect marginal costs via default risk and interest rate risk. Notably, I exploit variation in risk-weighted capital requirements both across lenders and across leverage levels within lender to identify the shadow cost of capital regulation. My empirical strategy therefore controls for differences across lenders that are common among products (lender shocks) and differences across products that are common across lenders (market shocks).

The estimated model captures several salient features of the data, such as market shares across lenders and LTVs, and performs well in predicting these features out of sample using additional data for 2017. On the demand side, I find an average loan demand elasticity of about 0.07 and a product demand elasticity of 5.9 across all mortgage products. These results straddle estimates from previous work based on different approaches (DeFusco and Paciorek (2017), Fuster and Zafar (2021)), as well as estimates from related work using structural approaches (Buchak et al. (2018a), Robles-Garcia (2019)).

On the supply side, I find an average markup of around 50 basis points, which corresponds to about 18% of the average interest rate in the data, and an average marginal cost of about 2.4%. Marginal costs are on average greater for mortgage products with higher LTV bands (capturing default risk) and a longer initial fixed period (capturing interest rate risk), which is intuitive. More importantly, I find that a 1-percentage-point higher risk-weighted capital requirement increases the average mortgage product marginal cost by 11% and interest rate by 10% (26 basis points).

Next, I use the estimated structural model to investigate the equilibrium effects of counterfactual leverage regulations in the mortgage market. Holding the estimated demand and marginal cost parameters fixed, I simulate different leverage regulations, allowing lenders and borrowers to best respond to the new regulations. These counterfactual simulations allow me to explore the design and equilibrium effects of alternative policies that are not observed in the data on mortgage rates, market structure, and risk.

In a first set of counterfactuals, I study the effects of removing the regulatory-risk-weight gap between large banks and other competitors. This analysis is motivated by recent proposals to reform capital requirements (Basel Committee on Banking Supervision (2016a, 2016b)). Most notably, I simulate a counterfactual in which all lenders calculate risk-weighted capital requirements using the standardized approach and an alternative counterfactual in which small and medium lenders also adopt a representative IRB model.

Imposing a common standardized approach to risk weights increases the equity buffer of larger lenders as well as their costs, which are passed on to borrowers with large changes in market shares and decline in both lender profits and consumer surplus. The provision of a representative IRB model to small lenders addresses the competitive distortion due to the differential

regulatory treatment, but with limited impact to consumer surplus, lenders profits, ex-post mortgage defaults, and the resilience of largest lenders. Overall, removing the policy-driven heterogeneity in risk weights reduces market concentration by approximately 20%. This procompetitive effect decreases the intermediation of the largest lenders, which played a central role in the crisis, and reduces their systemic importance.

Finally, I use the estimated model to explore possible interactions between capital requirements and limits to household leverage that have recently been discussed and implemented in some countries (Consumer Financial Protection Bureau (2013), Bank of England (2014), Acharya et al. (2020), DeFusco, Johnson, and Mondragon (2020)). I introduce a maximum LTV limit that removes from borrowers' menu all mortgages with leverage larger than 90%, both in an economy with risk-weighted capital requirements and in a counterfactual economy with homogeneous capital requirements (as was the case before the financial crisis under Basel I).

I find that a regulation targeting high LTV can reduce borrower defaults, which decline by 3% to 5%, but has a negative impact on originations and consumer surplus, as first-time buyers value mortgages with high leverage. Most notably, I show that the interactions of regulations on borrowers and lenders can have distributional effects and unintended consequences. Abolishing high-LTV mortgages has larger effects on consumer surplus and lender profits under homogeneous capital requirements, when low-LTV borrowers are "subsidizing" high-LTV borrowers, while the opposite is true under risk-weighted capital requirements. Finally, I show that introducing leverage limits in a market with risk-weighted capital regulation reduces large lenders' equity buffer and as a result increases systemic risk. This finding highlights the importance of looking at equilibrium effects, especially when multiple regulations are in place and the implications of changing a particular regulation may depend on other policies.

Related literature. This paper contributes to several strands of literature. First, I provide a new framework to study households' mortgage demand, which complements existing approaches in household finance using life-cycle portfolio choice models (Campbell and Cocco (2003), Campbell (2013)) or clever empirical identification strategies (DeFusco and Paciorek (2017), Best et al. (2020), Fuster and Zafar (2021)).⁴ Specifically, my structural approach uses state-of-the-art techniques from the industrial organization literature on differentiated product demand systems and on discrete-continuous choice models (Lancaster (1979), Dubin and McFadden (1984), Berry, Levinsohn, and Pakes (1995), Hendel (1999), Nevo (2001), Thomassen et al. (2017)). The characteristics approach captures rich heterogeneity in household preferences and product availability along several dimensions, which are otherwise difficult to model together. The discrete-continuous approach allows me to decompose the impact of interest rates on households' joint choice of lender, leverage, and

⁴ Gomes, Haliassos, and Ramadorai (2020) provide a recent and comprehensive survey of different approaches to mortgage choice and household finance in general.

loan size. Within the household finance literature, my paper is one of the first to also account for lenders' response to demand preferences using a structural equilibrium model.

Second, my work is related to a growing literature that employs structural techniques to quantitatively study the industrial organization of financial markets. Recent papers look at retail deposits (Egan, Hortaçsu, and Matvos (2017), Xiao (2020)), insurance (Kojien and Yogo (2016)), corporate lending (Crawford, Pavanini, and Schivardi (2018)), pensions (Hastings, Hortaçsu, and Syverson (2017)), credit cards (Nelson (2019)), and mortgages (Guiso et al. (2017), Buchak et al. (2018a, 2018b)). My paper is one of the first to apply similar techniques to the mortgage market and study the implications of leverage regulation for consumers and market structure. Most notably, while previous studies focus largely on a "representative" product for each lender and only model the choice across lenders, I exploit more granular variation in risk weights across asset classes *within* lenders to identify the elasticity of both product and loan demand and the impact of leverage regulation. In this sense, my paper is closest to Buchak et al. (2018a), who study several regulations in the United States using a structural industrial organization model that allows for rich heterogeneity in borrowers' product and quantity choices and lenders' financing options. My work complements their insights on supply-side factors that affect the equilibrium in the mortgage market by looking at the U.K. context, where banks keep most mortgages on their balance sheets and government interventions are less ubiquitous.⁵

Third, this paper contributes to the literature on the costs and benefits of capital regulations (Freixas and Rochet (2008), Admati and Hellwig (2014), Vives (2016), Greenwood et al. (2017)). Motivated by the global financial crisis, several papers have exploited different sources of variation to quantify the effects of capital requirements on the cost of capital and credit supply (Kisin and Manela (2016), Kovner and Van Tassel (2018), Gropp et al. (2019)); and to explore possible unintended consequences, such as risk taking (Acharya and Steffen (2015)), procyclicality (Behn, Haselmann, and Wachtel (2016a), Jiménez et al. (2017)), and regulatory arbitrage (Acharya, Schnabl, and Suarez (2013), Acharya, Engle, and Pierret (2014), Behn, Haselmann, and Vig (2016b), Plosser and Santos (2018)). On the empirical side, I provide new descriptive evidence of how risk-weighted regulation affects mortgage rates, with implications for lenders' competitiveness and specialization in different risk segments.

⁵ In the United States, for example, lenders securitize a large fraction of their mortgages via government-sponsored enterprises (GSEs), and the Federal Housing Administration provides government insurance for certain mortgage products with high leverage, thus affecting the pricing and origination incentives of private lenders. Australia and other European countries resemble the U.K. context more in terms of less government intervention. For an extensive discussion of differences in regulations and institutional characteristics across mortgage markets around the globe, see Lea (2010) and Seru (2019). Finally, much of the work on the supply-side aspects of the mortgage market focuses on the United States and Canada. Examining other countries can therefore provide new insights, in line with Badarınza, Campbell, and Ramadorai (2016)'s call for an international comparative approach to household finance.

My work is most related to recent papers studying the effects of risk-weighted capital requirements on credit supply in the corporate loan market. For example, Gropp et al. (2019) find that banks exposed to higher capital requirements reduced their credit supply of syndicated loans relative to unaffected banks; and Behn, Haselmann, and Vig (2016b) show that the introduction of model-based regulation in Germany led to lower risk weights and underprediction of defaults but higher interest rates and loan losses for large banks relative to smaller banks not adopting the model-based approach. I complement these papers by focusing on the mortgage market and combining a structural model with an identification strategy that addresses the challenge of separately identifying demand- and supply-side factors in a context in which multiple banking relationships are rare. Most notably, the structural approach allows me to evaluate equilibrium trade-offs of counterfactual policies, which can inform the policy debate when several alternatives are under consideration and experimentation can be costly.

My work also speaks to an alternative set of borrower-based macroprudential policies that have been implemented in several countries since the global financial crisis (Hanson, Kashyap, and Stein (2011), Cerutti, Claessens, and Laeven (2017)). Among others, DeFusco, Johnson, and Mondragon (2020) show that the Dodd-Frank “Ability-to-Repay” rule in the United States reduced household leverage, by increasing the cost of originating high-leverage mortgages, but with little effect on defaults, while Acharya et al. (2020) find that the introduction of LTI and LTV limits in Ireland led to reallocation across households with limited aggregate effects. Similar to these studies, I provide evidence on the effects of regulation on borrowers leverage, studying with administrative-level data a policy in the United Kingdom that limits the fraction of high-LTI mortgages that banks can originate. Different from previous studies, I develop a structural model that allows for rich heterogeneity across households and exploits detailed microdata for estimation. Most importantly, the model allows me to explore how borrower-based macroprudential policies interact with lender-based macroprudential policies, thus providing new insights on complex equilibrium interactions in the presence of multiple regulations.

Finally, my work is related to the recent macrofinance literature that has developed quantitative general equilibrium models of mortgage markets with heterogeneous agents (Greenwald (2018), Corbae and D’Erasmus (2019), Justiniano, Primiceri, and Tambalotti (2019), Begenau and Landvoigt (2021)). I adopt a partial equilibrium approach and focus mainly on the cost side of leverage regulation, which prevents me from implementing a fully fledged cost-benefit analysis of alternative regulations. Nevertheless, I explicitly model the interaction between leverage regulation and the competitive environment and the resulting implications for the pass-through of capital requirements to mortgage rates. In this sense, my work is also related to recent studies emphasizing the role of lenders’ market power for the transmission of policy interventions (Scharfstein and Sunderam (2016), Agarwal et al. (2017), Drechsler, Savov, and Schnabl (2017)). Most importantly, I show that market structure is an

endogenous outcome of leverage regulations that affect policy transmission to the real economy. This equilibrium interaction between leverage regulation and market structure has important implications for the design of policies to promote both stability and competition in the banking industry.

Overview. The remainder of the paper is organized as follows. Section I describes the institutional setting and data sources. Section II provides motivating facts on the U.K. mortgage market. Section III develops the demand and supply model. Section IV describes the estimation approach and the identification strategy. Section V discusses the main results. Section VI presents counterfactual analyses. Section VII concludes.

I. Institutional Setting and Data

A. Institutional Setting

While mortgages are extremely important in all countries, mortgage systems vary widely across countries (Campbell (2013)). Three key institutional features differentiate the U.K. mortgage market from mortgage markets in the United States, Canada, and other countries in continental Europe.

First, in contrast for example to the U.S. mortgage market, long-term fixed-rate mortgages are unavailable in the United Kingdom (Badarinza, Campbell, and Ramadorai (2017), Cloyne et al. (2019)). The typical mortgage contract is characterized by a 25-year amortization period but a relatively low interest rate for a short initial period, which is followed by a penalizing reset rate. The initial rate typically has a duration of two to five years and may be either fixed or floating. During the initial period, most contracts have large early repayment charges, typically 5% or 10% of the outstanding loan, which makes refinancing before the end of the initial period costly.⁶ After the end of the initial period, the mortgage rate resets to a standard variable rate (SVR). The SVR is always a floating rate and it is paid until either the borrower refinances the loan or maturity. Given the high prepayment penalty and the large jump in the reset rate, most borrowers in the U.K. refinance around the time when the initial period ends, which leads remortgaging to be a relatively frequent event in this market (Cloyne et al. (2019), Belgibayeva et al. (2020)).

Second, unlike other countries such as the United States and Canada, in the United Kingdom, there is limited consumer-based pricing or negotiation between the borrower and the lender (Allen, Clark, and Houde (2014), Buchak et al. (2018a)). As a result, the advertised rate is close to the rate that the borrower pays.⁷ Figure 1 shows a snapshot from a popular search platform for

⁶ A similar mortgage structure is common in Canada, where most mortgage contracts have a 25-year amortization period, but an initial term of between 1 and 10 years, during which time the interest rate can be either fixed or variable and the prepayment penalty is high (Allen, Clark, and Houde (2014)).

⁷ As the search platform Moneyfacts reports, “A personal Annual Percentage Rate is what you will pay. For a mortgage this will be the same as the advertised APR, as with a mortgage you can either have it or you can’t. If you can have the mortgage, the rate doesn’t change depending on

Provider	Initial monthly cost ▼	Initial rate ▼	Type of mortgage	Max LTV	Product fees	Overall cost for comparison ▼
HSBC 	£ 635.05	1.99% then 3.94%	Fixed for 25 months	90%	Yes	3.8% APRC representative
 The Loughborough Building Society	£ 639.44	2.05% then 4.99%	Discounted Variable for 24 months	90%	Yes	4.6% APRC representative
 YORKSHIRE BUILDING SOCIETY	£ 641.64	2.08% then 4.99%	Fixed for 25 months	90%	Yes	4.7% APRC representative
 Leek United BUILDING SOCIETY	£ 642.37	2.09% then 5.19%	Fixed for 27 months	90%	Yes	4.7% APRC representative
 first direct	£ 646.05	2.14% then 3.69%	Fixed for 24 months	90%	Yes	3.6% APRC representative

Figure 1. Choice set. This figure shows a snapshot from <https://www.moneysupermarket.com/mortgages/> of the five cheapest mortgages offered in a market after filling in information about the value of the property and the loan amount. (Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/jofi.13072))

mortgages in the United Kingdom after filling in information on the value of the property and the loan amount. The key mortgage characteristics are the loan provider, the type of interest rate, and the maximum LTV (e.g., a mortgage offered by Barclays with a two-year fixed rate and a maximum 90% LTV). I define a product as a lender, interest rate type, and maximum LTV triplet and I regress loan-level interest rates on product fixed effects and an extensive set of additional controls.

Panel (A) of Figure IA.1 in the [Internet Appendix](#) shows that my product definition alone captures more than 70% of the variation in the loan-level rate. The R^2 reaches 85% when I interact the product dummies with time dummies, and reaches more than 90% when I also include origination fee dummies. Adding dummies for the location of the house and borrower-level controls (age, income, house value, joint application, employment status) does not explain the residual variation in the rate. While I do not have access to credit score data and therefore cannot explicitly control for the effect of borrower credit scores on mortgage rates, Robles-Garcia (2019) shows that they do not seem to matter for mortgage pricing in the United Kingdom after conditioning for the relevant product characteristics.⁸ This also differentiates the United Kingdom

your credit score, which it may do with a credit card or a loan” (source: <https://moneyfacts.co.uk/guides/credit-cards/what-is-an-apr240211/>).

⁸ Credit scores may play a role in mortgage approval in the United Kingdom, but unfortunately, I do not have access to application and rejection data that together with credit score data, would be needed to explore this additional dimension. For this reason, I focus on mortgage pricing. The remaining variation in pricing in Figure IA.1 may be due to two possible factors. First, unobservable product characteristics. Even if I control for the main factors affecting price, lenders may use some other unobservable product characteristics to segment the market (e.g., cash-back offers).

from the United States, where credit scores together with LTV bands affect mortgage rates.⁹

Third, the industrial organization of the mortgage market in the United Kingdom has some distinctive features. In particular, the U.K. banking market has become concentrated as a result of mergers and acquisitions over the last 40 years, particularly after the global financial crisis. As a result, in 2015, the six largest lenders accounted for more than 75% of new mortgage originations.¹⁰ In more recent years, competition has increased with the entry of new challenger banks and with the introduction of regulatory interventions, such as forced divestment, but the market remains concentrated by international standards.¹¹ Thus, the industrial organization of the mortgage market in the United Kingdom is more similar to that of countries like Canada, where a few large banks account for the majority of mortgage originations, than to the United States (Allen, Clark, and Houde (2014), Scharfstein and Sunderam (2016)).

B. Data

My paper combines several sources of administrative data. The main data set is the Product Sales Database (PSD) on residential mortgage originations collected by the Financial Conduct Authority (FCA). These data include the universe of residential mortgage originations by regulated entities since 2005.¹² I observe the main contract characteristics of the loan (rate type, repayment type, initial period, interest rate, lender), the borrowers (income, age), and the property (value, location, size). For the structural estimation, I focus on the years 2015 and 2016, during which all lenders report all contract characteristics that I employ in the analysis.

I complement the information on mortgage originations with four additional data sets. First, I use an additional source collected by the FCA that contains information from lenders' balance sheets on the performance of outstanding

Second, while I observe the date when the borrower gets a mortgage, I do not know when exactly the deal was agreed upon and thus the time dummies capture variation in prices imperfectly.

⁹ In the United States, both LTV and credit scores enter the loan-level price adjustment (LLPA) grids that provide the additional guarantee fees that lenders must pay government agencies (e.g., Fannie Mae) for guaranteeing the mortgage. These fees are then passed on to borrowers via a higher interest rate to pay for credit risk.

¹⁰ The six largest lenders (the so-called "big six") are Barclays, HSBC, Lloyds Banking Group, Nationwide Building Society, Royal Bank of Scotland, and Santander United Kingdom. The other main lenders in the mortgage market are building societies and challenger banks. Building societies are owned entirely by their members and mainly provide saving and mortgage lending services to their members, and thus resemble credit unions in the United States.

¹¹ Metro Bank—a challenger bank—was granted a license in 2010 and represents the first high-street bank to be granted such a license in over 150 years. Similarly, in 2009, the European Commission forced the largest U.K. bank—Lloyds TSB—to divest some of its activities to a new entity—TSB Bank—to increase competition in the U.K. banking sector.

¹² The FCA PSD focus on regulated mortgage contracts only and therefore exclude other regulated home finance products, such as home purchase plans and home reversions, and unregulated products, such as second charge lending and buy-to-let mortgages.

mortgages as of June 2016. Second, I exploit data on lenders' capital requirement and resources from the historical regulatory databases held by the Bank of England together with additional information from a survey of all lenders adopting IRB models in the United Kingdom on risk weights applied to mortgages by LTV band.¹³ Third, for all lenders in my sample, I collect postcode-level data on their branches in the United Kingdom in 2015 from SNL financial. Fourth, I match borrowers' house with the house price index at the postcode level from the Office for National Statistics (ONS) database.

Panel A of Table I reports summary statistics for the universe of mortgages originated in the United Kingdom in 2015 and 2016 with an LTV above 50%.¹⁴ In Panel A, I show the main data set on mortgage originations. I observe more than 1 million mortgage contracts, with an average rate of about 2.7 percentage points and an origination fee of £600. Mortgages fixed for two and five years together account for more than 85% of all originations. The average loan value is about £170,000, with an LTV of 77% and an LTI of 3.3. The sample is balanced across first-time buyers, home movers, and remortgagers. The average maturity is 25 years, and the average borrower is 35 years old with an income of around £57,000.

In Panel B of Table I, I report summary statistics on lenders' capital requirements, risk weights, and branches. The capital requirements include both minimum requirements under Basel II (Pillar I, or 8% of risk-weighted assets), as well as lender-specific supervisory add-ons (Pillar II). Total capital resources include all classes of regulatory capital, including common equity Tier 1, additional Tier 1, and Tier 2. The average capital divided by total risk-weighted assets is 17% when I focus only on Tier 1 and 21% when I include all classes of regulatory capital. The average capital requirement is 12%, ranging from a minimum requirement of 8% to a maximum of 22%, including all add-ons. The average risk weight is 17% and there is a lot of variation across lenders, leverage levels, and over time: the standard deviation is 11% and risk weights vary from a minimum of less than 1% to a maximum of almost 70%. The average number of lenders' branches in a postcode area is about seven, ranging from a minimum of 1 to a maximum of 63.

Finally, Panel C of Table I reports statistics on mortgage performance as of June 2016. On average 1.3% of outstanding mortgages are in arrears, which is defined as 90 or more days delinquent on monthly payments. Almost 85% of mortgages are not paying the SVR, which, on average, is around 3.7%.

¹³ The data on capital requirements are standard and have been used in previous studies (De Ramon, Francis, and Milonas (2016), Harimohan, McLeay, and Young (2016)). The information on risk weights was collected by the Bank of England and the Competition and Market Authority to study the effects of the change from Basel I to Basel II on mortgage prices and is described in detail in Benetton et al. (2020).

¹⁴ My analysis focuses on leverage regulation and risk, so I exclude all mortgage transactions in which borrowers have more than 50% of their equity in the house. These are mainly remortgagers with a probability of being in arrears below 0.1%.

Table I
Summary Statistics

The table reports summary statistics for the main variables used in the analysis. In Panel A, I show the main variables used in the analysis based on the universe of mortgages originated in the United Kingdom in 2015 and 2016 with an LTV above 50%. Interest rate is the interest rate at origination expressed in percentage points; fee is the origination fee in pounds; fix two (five) years is a dummy for products with an initial period of two (five) years; loan value is the loan amount borrowed in thousand pounds; first-time buyers, home movers, and remortgagers are borrower-type dummies; maturity is the original maturity of the mortgage in years; gross income is the original gross income in thousand pounds; and age is the age of the borrowers in years. In Panel B, I present results for lender variables. The capital requirements include both minimum requirements under Basel II (Pillar I, or 8% of risk-weighted assets) as well as lender-specific supervisory add-ons (Pillar II). Total capital resources include all classes of regulatory capital, including common equity Tier 1, additional Tier 1, and Tier 2. I report them as a percentage of total risk-weighted assets. Risk weights are expressed in percentage points. Branches is the number of branches for each lender in each postcode area. In Panel C, I summarize data on mortgage performance. Arrears is the fraction of outstanding mortgages in arrears as of June 2016. Not paying SVR is the fraction of mortgages not paying a standard variable rate (SVR) as of June 2016. SVR is the standard variable rate expressed in percentage points.

	Mean	SD	Min	Max
Panel A: Loan-Borrower				
Interest Rate (%)	2.65	0.81	1.24	5.19
Fee (£)	631.50	602.25	0.00	2,381.00
Fix 2 years	0.64	0.48	0.00	1.00
Fix 5 years	0.22	0.42	0.00	1.00
Loan value (£.000)	177.89	93.94	47.99	631.29
LTV (%)	77.60	12.16	50.00	95.00
LTI	3.32	0.89	1.09	5.00
First-time buyers	0.34	0.47	0.00	1.00
Home movers	0.32	0.47	0.00	1.00
Remortgagers	0.32	0.47	0.00	1.00
Maturity (Years)	25.69	6.64	5.00	40.00
Gross income (£.000)	57.08	30.86	16.79	233.72
Age (Years)	35.73	8.15	17.00	73.00
Panel B: Lender				
Capital ratio tier 1 (%)	17.24	7.19	6.93	43.50
Capital ratio total (%)	21.11	6.73	9.90	44.20
Capital requirement (%)	12.03	2.40	8.00	22.54
Risk weights (%)	16.95	11.21	0.87	68.05
Branches (Number)	6.90	7.10	1.00	63.00
Panel C: Performances				
Arrears (%)	1.32	11.42	0	100
Not paying SVR (%)	84.66	36.02	0	100
Standard variable rate (SVR) (%)	3.71	1.01	2.25	4.74

II. Motivating Facts

The U.K. mortgage market has several institutional features that make it an excellent laboratory for investigating the relationship between leverage regulation, mortgage pricing, and market structure. In this section, I provide a brief overview of four empirical facts that guide my assumptions underlying the structural model of Section III, and I describe some of the main sources of variation used in the identification strategy described in Section IV.

A. Leverage Regulation

The U.K. market has been at the forefront of policies to regulate both lenders and households' leverage. On the lender side, since the 1990s the U.K. microprudential regulator—the Financial Services Authority—has imposed time-varying, bank-specific minimum capital requirements on the banks under its purview (Aiyar et al. (2014)). In addition, since 2008 two approaches to calculate risk weights have coexisted: the standardized approach and the IRB approach. Under IRB, banks' own internal models are used to estimate risk components such as probabilities of default and loss given default, which are then used as inputs in the risk-weight functions defined by regulators. The six largest lenders in the United Kingdom have all adopted IRB models since 2008 when the capital regulation changed from Basel I to Basel II. In contrast, with very few exceptions, medium and small lenders have opted for the standardized approach, mostly because of the large fixed compliance cost associated with IRB models (Competition and Markets Authority (2015), Benetton et al. (2020)).

As a result, a two-tier system has prevailed, with large variation in risk weights for potentially *similar* mortgage products.¹⁵ Figure 2 shows risk weights for U.K. lenders in 2015 as a function of LTV. For lenders adopting the standardized approach, risk weights are fixed at 35% for LTVs up to 80%, and they increase to 45% on incremental balances above 80%. In contrast, lenders adopting an IRB model have risk weights that increase with LTV along the entire distribution. The average gap between the average IRB risk weight and the risk weight under the standardized approach is about 30 percentage points for mortgages with an LTV below 50%, compared to less than 15 percentage points for mortgages with leverage above 80%.

An alternative way to regulate leverage in the mortgage market is to impose explicit limits on how much households can borrow. Among recent examples of such policies are maximum debt-to-income (DTI) limits in the United States (DeFusco, Johnson, and Mondragon (2020)), and both LTI and LTV limits in Ireland (Acharya et al. (2020)). In the United Kingdom, in June 2014, the Financial Policy Committee recommended that regulated lenders limit mortgage

¹⁵ In Section II.D, I explore borrowers' selection looking at ex-ante borrower characteristics and ex-post mortgage performance.

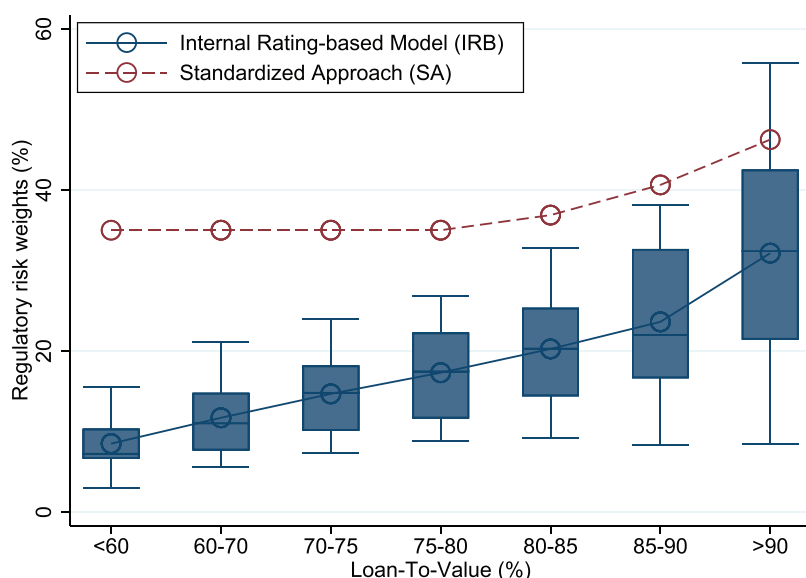


Figure 2. Risk weights across lenders and LTVs. This figure shows the average risk weights for two groups of lenders at different LTVs. IRB includes all lenders in the sample used for the structural estimation that adopt an IRB model in calculating capital requirements. The internal model of the lender is subject to supervisory approval. The distribution of IRB risk weights within each LTV band is represented by a Tukey boxplot, where the box represents the interquartile range (IQR) and the whiskers represent the most extreme observations still within $1.5 \times \text{IQR}$ of the upper/lower quartiles. SA includes all lenders in the sample used for the structural estimation that adopt the standardized approach. For the latter group, the risk weights are set by the regulator in a homogeneous manner across banks and vary between 35% and 45% based on the LTV of the loan. (Color figure can be viewed at wileyonlinelibrary.com)

originations with an LTI above 4.5% to 15% of the total number of new mortgage loans (Bank of England (2014)).¹⁶

In Section II of the [Internet Appendix](#), I provide evidence on the effect of the policy on mortgage originations in a difference-in-difference setting, exploiting variation before and after the policy and across lenders in the fraction of mortgages with an LTI higher than 4.5 before the policy. I find that more affected lenders reduce their fraction of high-LTI mortgages by almost 4% more relative to other lenders, with the effect stronger in the market for first-time buyers.

In practice, both risk-weighted capital requirements for lenders and maximum LTV or LTI for households are complementary tools to regulate the

¹⁶ For more details about the recommendation, see <http://www.bankofengland.co.uk/financialstability/Pages/fpc/loanincome.aspx>. The main statement reads as follows: “The Prudential Regulation Authority (PRA) and the Financial Conduct Authority (FCA) should ensure that mortgage lenders do not extend more than 15 percent of their total number of new residential mortgages at loan to income ratios at or greater than 4.5. This recommendation applies to all lenders that extend residential mortgage lending in excess of £100 million per annum. The recommendation should be implemented as soon as is practicable.”

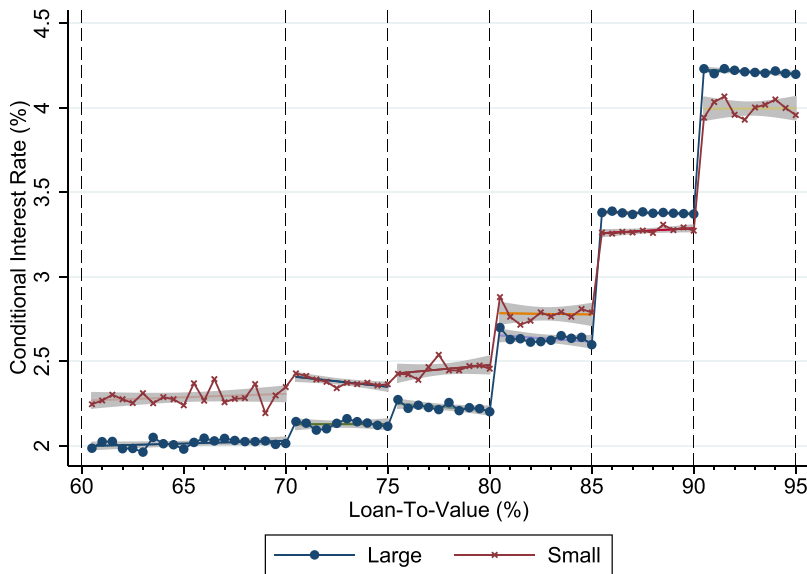


Figure 3. Pricing risk and regulation. This figure shows the conditional interest rate from the regression $r_j = controls_j + \sum_{j=60}^{95} ltv_j + \epsilon_j$, where $controls_j$ are fixed effects for time, interest rate type, and lenders and ltv_j are 0.5 percentage points LTV bins. The dotted vertical lines denote the maximum LTV of 60%, 70%, 75%, 80%, 85%, 90%, and 95%. The blue and red lines plot the schedule for a representative large lender adopting the IRB model and a representative small lender using the standardized approach. (Color figure can be viewed at wileyonlinelibrary.com)

mortgage market. In this paper, I provide a framework to study borrower- and lender-based macroprudential policies and their interactions, which can have important equilibrium effects. Most notably, in the counterfactual analysis of Section VI, I explore two types of regulations: risk-weighted capital requirements for lenders and maximum LTV limits for borrowers.

B. Mortgage Pricing

Do differences in leverage regulation affect mortgage pricing? In this section, I discuss evidence on the relation between risk-weighted capital requirements and mortgage rates.¹⁷ First, in Figure 3, I explore the variation in rates across LTVs and lenders. I plot the mean predicted interest rate as a function of the LTV obtained from a regression of loan-level interest rates on a step function with steps at each 0.5 percentage point LTV bin and a rich set of mortgage, lender, and macro controls. I focus on a representative large lender using the IRB model and a representative small lender using the standardized approach. Both lenders set the interest rate as an increasing schedule of the LTV, which captures default risk (Schwartz and Torous (1989), Campbell and

¹⁷ In Section II of the [Internet Appendix](#), I discuss additional evidence on the effect of risk weights and risk-weighted capital requirements on mortgage rates.

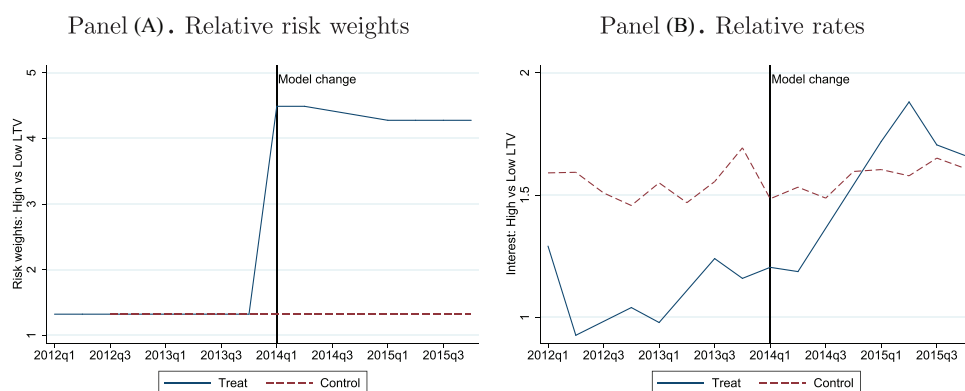


Figure 4. Model change. Panel (A) shows the relative risk-weight, while Panel (B) shows the relative interest rate of a maximum LTV of 95% relative to a maximum LTV of 70% for a treated lender before and after the adoption of an IRB model and a control lender that continues to adopt the standardized approach. (Color figure can be viewed at wileyonlinelibrary.com)

Cocco (2015)), with discrete jumps at certain maximum LTV thresholds as already documented by Best et al. (2020).¹⁸ The rate schedule of the large lender shows clear discontinuous jumps at maximum LTVs, while the small lender increases sharply the rate only for LTVs above 80%, when risk weights start to increase as shown in Figure 2. The large lender offers a more competitive interest for low-LTV mortgages. The gap in prices closes for intermediate-LTV mortgages and even reverses for products with an LTV above 85%, consistent with the shrinking gap in risk weights between internal models and the standardized approach.

Second, I further explore the relation between regulation and pricing by exploiting *within* lender variation in risk weights and interest rates over time. One concern with the pattern displayed in Figure 3 is that there are unobservable factors varying jointly across lenders and LTVs that affect pricing, and could also be correlated with risk weights. To mitigate this concern, I exploit the approval of an IRB model for a medium-sized lender and compare its risk weights and mortgage rates to those of another medium-sized lender that continues to adopt the standardized approach.

Figure 4 plots the average risk-weight and the average interest rate for the two medium-sized lenders for mortgages with a maximum LTV of 95% relative to those with a maximum LTV of 70%.¹⁹ Panel (A) shows a large increase for the lender adopting the IRB model in the risk-weight of high-LTV mortgages,

¹⁸ In Panel (B) of Figure IA.1 in the Internet Appendix, I reproduce the same figure separately for mortgages with different initial periods. I find that for a given LTV, mortgages with a longer fixed rate are more expensive than mortgages with a shorter fixed rate. This finding is consistent with higher refinancing risk embedded in a contract with a longer fixed duration (Deng, Quigley, and Order (2000), Rose (2013), Beltratti, Benetton, and Gavazza (2017)).

¹⁹ Figure IA.5 in the Internet Appendix plots the levels of the average risk weights and mortgage rates for the lenders in this event study.

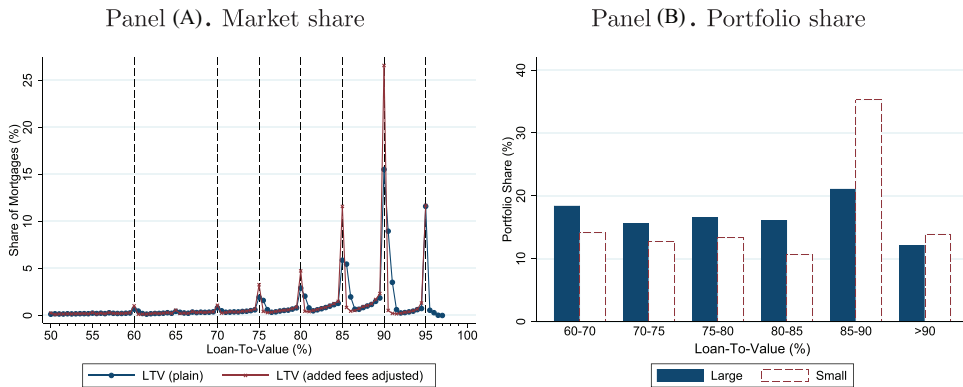


Figure 5. Originations across LTVs. Panel (A) shows the share of mortgages originated at different LTV bins for first-time buyers. Each bin is 0.5 percentage point wide. The blue line shows the plain distribution where the LTV is computed as the ratio between the loan value divided by the house value. The red line controls for fees by subtracting the fees added to the loan from the loan value. The dotted vertical lines denote maximum LTVs of 60%, 70%, 75%, 80%, 85%, 90%, and 95%. Panel (B) shows the portfolio shares for a representative large lender adopting the IRB model and for a representative small lender using the standardized approach. (Color figure can be viewed at wileyonlinelibrary.com)

which makes the relative risk-weight jump from about 1.5 to more than 4. Panel (B) shows that the average interest rate on high-LTV mortgages relative to low-LTV mortgages increases from about 1.1 to about 1.6. At the same time, the lender that continues to adopt the standardized approach shows no change in relative pricing.

C. Mortgage Originations

I next provide evidence on equilibrium mortgage originations across LTVs and lenders, which are the two main dimensions of variation in leverage regulation. Figure 5, Panel (A), shows the LTV choice of first-time buyers. The vast majority of first-time buyers are concentrated at high maximum LTVs, with more than 25% borrowing (almost) exactly 90% of the value of their house.²⁰ This bunching behavior allows me to model the leverage choice as a discrete choice of the LTV band. Figure 5, Panel (B), presents the quantity counterpart of the analysis of prices in Figure 3. Most notably, I compute the portfolio shares of a representative large lender using an IRB model and a representative small lender using the standardized approach. The large-lender portfolio is evenly distributed across all leverage levels, while the small lender issues most mortgages at high LTVs, which is the segment in which its relative disadvantage is lower.

²⁰ Best et al. (2020) show a similar but less pronounced pattern for remortgagers. The more pronounced pattern that I find for first-time buyers may be due to the fact that they do not have the flexibility that equity extraction may offer remortgagers.

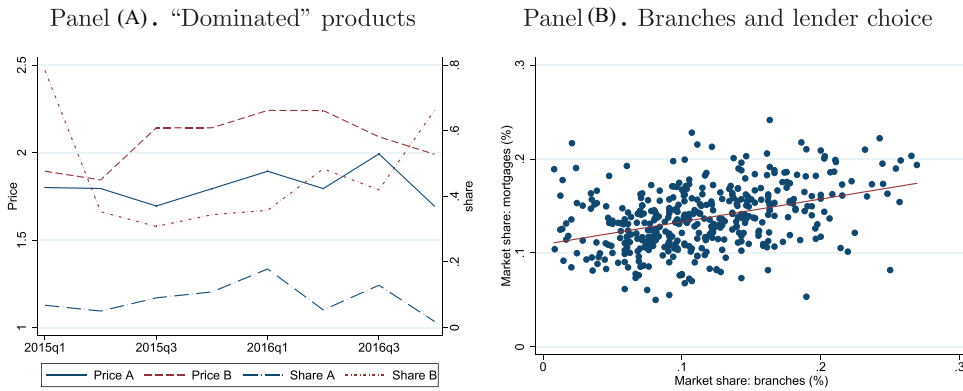


Figure 6. Originations across lenders. Panel (A) shows the price and market share of two products for first-time buyers offered by two different lenders with the same initial period (two years), the same maximum LTV (60%) and similar quantities (£140,000 to 160,000). The price is the full APR, which includes the initial interest rate and origination fees. The market share is computed as the fraction of people who buy a given product in a specific quarter over the total number of mortgage borrowers in that quarter. Panel (B) shows the correlation between the share of branches and the share of mortgages across postcode areas in the United Kingdom for the six largest lenders. (Color figure can be viewed at wileyonlinelibrary.com)

Overall, the evidence so far suggests that differences in leverage regulations across lenders and products can affect mortgage pricing and originations. The ultimate impact on market structure, however, may also depend on other factors that differ across lenders, such as incumbent advantage and brand recognition, which can affect borrower elasticities to the interest rate and substitution patterns following changes in regulation. I explore some of these factors further in Figure 6.

Figure 6, Panel (A), plots the price and market share of two mortgage products with the same maximum LTV (60%), interest rate type (two years fixed), and loan size (£140,000 to 160,000) but that are offered by two different lenders. I find that the mortgage with the higher price has the higher market share for the full period under analysis. In my model, I account for factors (e.g., brand value) that can help explain this counterintuitive effect and I further account for the role of lenders' branch network in borrowers' choices.²¹ Figure 6, Panel (B), shows the correlation for the six largest lenders between the branch share and the mortgage share in each postcode area. I find a strong positive

²¹ A possible explanation also comes from the supply side, with low price—low market share products approved only for some customers. Due to data limitations, I cannot test this hypothesis, but given the low leverage (60%), rejections are less likely to be a concern. I do not have information on lenders' approval decisions, so I assume that all borrowers of a certain type have access to the advertised rate and take the best alternative. To limit concerns about rejection, I restrict the choice set based on observable borrower characteristics and affordability criteria as explained in Section IV.A.

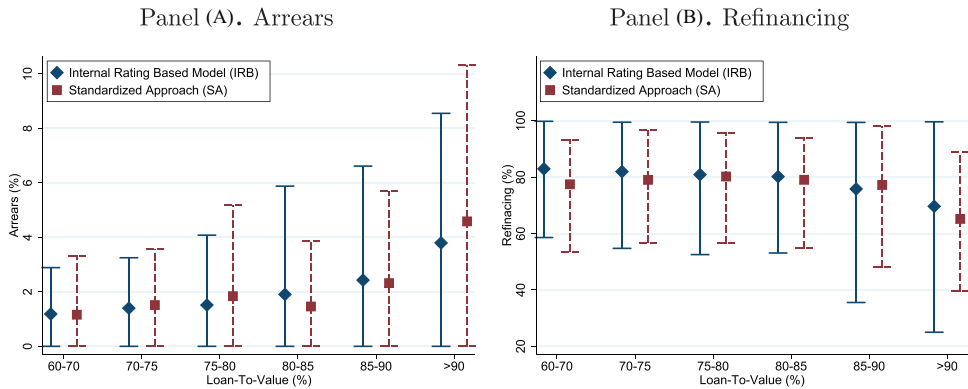


Figure 7. Performance across lenders and LTVs. Panel (A) shows the mean, 5th percentile, and 95th percentile of the distribution of mortgages in arrears for different LTV brackets for lenders using an IRB model and for lenders using a standardized approach to compute risk weights. Panel (B) shows the mean, 5th percentile, and 95th percentile of the distribution of mortgages that are refinanced (i.e., not paying the SVR) for different LTV brackets for lenders using an IRB model and for lenders using a standardized approach to compute risk weights. (Color figure can be viewed at wileyonlinelibrary.com)

relationship between local branch presence and mortgage originations.²² Accounting for brand value and the branch network in the demand model is important to capture factors that can affect the demand elasticity (e.g., limited substitution due to local shopping), as the distance between the borrower and the lender continues to play an important role even in modern lending markets (Becker (2007), Scharfstein and Sunderam (2016), Nguyen (2019)).

D. Mortgage Performances

My last set of motivating empirical facts relates to mortgage performance across LTVs and lenders. I study two main risks in the mortgage markets: default and refinancing. For each product, I capture default as the fraction of mortgages originated since 2005 that are in arrears in 2016 out of the total number of mortgages on lenders' balance sheet, and I capture refinancing risk as the fraction of outstanding mortgages in 2016 that are on an SVR out of the total number of mortgages on the lenders' balance sheet. The refinancing variable is defined as one minus the share paying the SVR, which is the reset rate that borrowers pay at the end of the initial fixed or discounted period.

Figure 7 shows default and refinancing patterns, looking at the joint variation across LTVs and across lenders using different models to compute risk

²² Figure IA.2 in the Internet Appendix shows that lenders originate more mortgages in areas in which they have a large share of branches. To control for differences in nationwide popularity and in local market demand and branch networks, I also run a difference-in-difference specification with lender and market fixed effects. Figure IA.3 shows the results. I find that a lender has a 3% higher mortgage share in areas in which it is in the top quintile of the branch share distribution compared to areas in which it has no branches.

weights.²³ In Panel A, the fraction of mortgages in arrears increases monotonically with the maximum LTV, from 0.8% for mortgages with an LTV below 70%, to more than 3% for mortgages with an LTV above 90%. This pattern is reflected in the pricing schedules of Figure 3.²⁴ Interestingly, I do not find significant differences between large lenders using IRB models and medium small lenders using the standardized approach. In Panel B, I find a slight decrease in refinancing for high-LTV mortgages, but as before no differential patterns between lenders following different approaches to compute risk weights.

The absence of large differences in ex-post performance by LTV across lenders using different models to compute risk weights suggests that the large gaps in risk weights depicted in Figure 2 may be due at least in part to differences in regulatory regime and in bank practices, rather than to intrinsic differences in actual risk.²⁵ This comparison is suggestive, given that I only observe risk weights, ex-ante characteristics, and ex-post outcomes, as the exact internal models are proprietary. However, the existence of a possible mismatch between risk weights and realized risk or expected risk as proxied by ex-ante characteristics is consistent with previous evidence (Aiyar et al. (2014), Behn, Haselmann, and Vig (2016b), Plosser and Santos (2018)).²⁶ The comparison between risk weights and ex-post performance is especially relevant for low-LTV mortgages, where arrears are less likely and loss-given-default is lower as a result of the lower initial leverage. A conservative regulator that sets risk weights for the standardized approach together with large banks trying to economize on capital may generate a large gap in risk weights for mortgages that do not exhibit large differences in ex-post performance.

III. A Structural Model of the U.K. Mortgage Market

In this section, I develop a structural model of mortgage demand and pricing to study the equilibrium implications of changes in leverage regulation. I begin by specifying household utility as a function of mortgage characteristics and

²³ Figure IA.4 in the [Internet Appendix](#) compares borrower characteristics at origination, namely, age and income. I do not find significant differences between lenders and across LTVs in the demographics of the mortgage borrowers who they lend to.

²⁴ The increase in mortgages in arrears with the LTV can be due to both adverse selection, with more risky borrowers choosing higher-LTV mortgages, and moral hazard, because the higher rate increases the likelihood of default. While I cannot distinguish between these different channels, in the pricing model, I consider how lenders account for default risk when setting mortgage prices.

²⁵ For example, one aspect that can generate substantial variation in risk weights across banks adopting IRB models is the time horizon used to estimate the risk weights. Some banks used only the last year or so (point-in-time estimates), while others used much longer periods (through-the-cycle approach).

²⁶ Aiyar et al. (2014) show that in the U.K. regulatory, decisions about capital requirements are often based more on organization structures, systems, and reporting procedures than on credit risk factors. Plosser and Santos (2018) find that in the U.S. low-capital banks in loan syndicates bias risk estimates to improve their regulatory ratios. Behn, Haselmann, and Vig (2016b) show that in Germany, the introduction of model-based regulation leads to lower risk weights and underprediction of default for banks adopting internal models.

deriving both product and loan demand. I then develop a pricing equation that incorporates the empirical facts described in Section II.

A. Household Demand

In each market m and quarter t , I_{mt} heterogeneous borrowers, indexed by i , choose a mortgage to buy a house. I define a market m as a combination of geographic region and demographic characteristics.²⁷ Borrowers simultaneously choose a mortgage product indexed by j , which is defined as a lender, interest rate type, and maximum LTV triplet, and a loan amount, given their preferences and budget constraint. I follow the characteristics approach (Lancaster (1979)) and assume that (i) each mortgage can be represented as a bundle of attributes and (ii) borrowers have preferences over these attributes.

The indirect utility of household i selecting product j in market m and quarter t is given by

$$V_{ijmt} = \bar{V}_{ijmt}(Y_i, D_i, X_j, r_{jmt}, B_{ij(l)}, \zeta_i, \xi_{jmt}; \theta_m) + \varepsilon_{ijmt}, \quad (1)$$

where Y_i is household income, D_i is a vector of household demographics (e.g., age), r_{jmt} is the interest rate of product j in market m and quarter t , X_j is a vector of time invariant product characteristics (interest rate type, brand, maximum LTV), $B_{ij(l)}$ is lender l 's branch network in the location of household i , ζ_i captures borrowers' unobserved characteristics (e.g., wealth, risk-aversion, housing preferences), ξ_{jmt} captures unobservable characteristics of product j (e.g., advertising, screening) that affects the utility of all borrowers in market m at time t , ε_{ijmt} is an idiosyncratic taste shock, and θ_m is a vector that collects the demand parameters common to all borrowers in market m .²⁸

I construct the choice set comparing borrowers with similar observable characteristics in the same market and quarter, and I impose two additional restrictions based on affordability and liquidity constraints. First, households may not be able to borrow up to the desired leverage due to supply side restrictions (such as LTV or LTI limits). Second, liquidity constraints may limit the ability of the household to increase their downpayment and consider products with lower maximum leverage. Both types of constraints restrict the choice set of households in terms of the maximum LTV among the full set available in the market.

The borrower's constrained problem becomes

$$\max_{j \in J_i} V_{ijmt} = \bar{V}_{ijmt} + \varepsilon_{ijmt}, \text{ with } J_i \subseteq J_{mt}$$

²⁷ The regions are Scotland, Wales, North East, North West, Yorkshire and the Humber, West Midlands, East Midlands, South West, South East, East of England, and Greater London. In terms of demographics, I define four categories based on above- versus below-median income and age.

²⁸ I allow all of the structural parameters collected in the vector θ_m to vary across markets. Continuous variation in observable demographics (Y_i, D_i) and unobservable borrower characteristics (ζ_i) generate heterogeneity also within market across individual borrowers in preferences and demand elasticities.

$$j \in J_i \text{ if } j \in \{\max LTV_i - 1, \max LTV_i, \max LTV_i + 1\},$$

where J_{mt} is the total number of products available in market m and quarter t . In the benchmark case, the borrower has access to all products, so that $J_i \equiv J_{mt}$. I implement affordability constraints by (i) restricting the choice set of the borrower to products with the chosen maximum LTV and only one step above and below, and (ii) considering a representative product with the chosen observed maximum LTV as the outside option.²⁹

I assume that borrowers choose the mortgage product that gives them the highest utility among the products available to them. This assumption is particularly suitable for the mortgage market, in which the vast majority of borrowers take only one mortgage at a time. Thus, borrower i chooses product j if $V_{ijmt} > V_{ij'mt}$, $\forall j' \in J_i$. The probability that borrower i chooses product j in market m and quarter t given the value of his unobserved heterogeneity ζ_i therefore equals

$$p_{ijmt}(\zeta_i) = \text{Prob}(V_{ijmt} \geq V_{ij'mt}, \forall j' \in J_i). \quad (2)$$

Given the chosen product, the borrower decides the optimal quantity (q_{ijmt}), which I obtain using Roy's identity,

$$q_{ijmt} = -\frac{\frac{\partial V_{ijmt}}{\partial r_{jmt}}}{\frac{\partial V_{ijmt}}{\partial Y_i}} = q_{ijmt}(Y_i, D_i, X_j, r_{jmt}, \zeta_i, \xi_{jmt}; \theta_m). \quad (3)$$

The demand model jointly described by equations (2) and (3) captures borrowers' discrete product demand and continuous loan demand, respectively, in a general and flexible way. In practice, the model described so far implies only one exclusion restriction: branches enter the discrete product choice (equation (2)) but not the continuous quantity choice (equation (3)).³⁰

Branches as well as other product characteristics in X_j , such as the interest type and brand, allow for horizontal differentiation, which can lead to imperfect substitution across mortgages as a result of changes in relative interest rates. For example, higher branch presence can increase household utility because they lead to spatial differentiation. A large branch presence allows the household to walk into a branch when needed, lowering transaction costs. More branches can also make the lender more salient to the borrower by increasing the probability that the borrower will consider it. Moreover, in the absence of data on borrower assets, the local share of branches can proxy for preexisting relations between the borrower and the lender (e.g., current account).³¹

²⁹ I discuss the construction of the borrower-specific choice set in more detail in Section IV.A.

³⁰ Figure IA.3 in the Internet Appendix provides suggestive evidence that branches affect lenders' market share, but are not significantly related to differentially larger loan amounts. I discuss the identification assumptions in detail in Section IV.C.

³¹ In the U.K. mortgage market, borrowers search for mortgage products and apply via branches, intermediaries, and online comparison websites. The application process is long and

In the model, the interest rate r_{jmt} does not vary across individual i , in line with the lack of individual-based pricing discussed in Section I.A, but does vary across all characteristics defining products j , consistent, for example, with the pricing schedule across maximum LTVs and across lenders in Figure 3. The sensitivity to the interest rate is driven jointly by the discrete-continuous choice and varies based on differences in borrowers' characteristics and preferences both across and within markets as well as on differences in product characteristics.³²

Finally, I allow unobservable product heterogeneity (ξ_{jmt}) to affect the household mortgage and quantity choice. In the case of mortgage products, the unobservable term can capture "quality" as in standard industrial organization models (Berry, Levinsohn, and Pakes (1995), Nevo (2001)). Even if I observe the most relevant mortgage characteristics, lenders can offer optional services and contracts can include additional features (e.g., cash back, payment holidays) that may increase the "quality" of the product, *ceteris paribus*. Additionally, advertising plays an important role in the mortgage industry and could further justify the inclusion of unobservable product heterogeneity (Goeree (2008), Gurun, Matvos, and Seru (2016)).³³

B. Lender Pricing

In each market m and quarter t , L_{mt} lenders maximize (expected) profits by setting a price for each product they offer. I assume that lenders' main source of revenue is the net interest income from the monthly payments. In the data, I observe that U.K. lenders adopt national prices for identical products across geographic markets and borrower demographics, that is, $r_{jmt} = r_{jt}$. The present value of net interest income from a risk-free mortgage to borrower i with fixed

can be very costly. Ideally, I would like to observe the true household choice set when applying for mortgages, but this information is not available in most settings, especially for large administrative data sets. See Basten (2020) and Michelangeli and Sette (2016) for examples of settings in which the choice sets are observable.

³² In Section III of the Internet Appendix, I report the formulas employed to compute both discrete and continuous elasticities to the interest rate. Using Roy's identity requires that some parameters are common to the discrete-continuous choice. Alternative approaches to model the discrete-continuous problem in similar settings are Crawford, Pavanini, and Schivardi (2018) and Buchak et al. (2018a). Crawford, Pavanini, and Schivardi (2018) specify a separate loan demand function conditional on borrowing and adopt a control function approach to account for endogeneity in rates, while Buchak et al. (2018a) include the loan amount choice directly inside the indirect utility through a penalty term and identify it using a bunching technique.

³³ My working assumption is that borrowers have full knowledge of prices and loan characteristics of all mortgages in their choice set when they make their decision. The U.K. mortgage market has a large number of products and both the turnover rate and the frequency of price changes are high. However, detailed product information is readily available from price comparison websites and many borrowers use a broker to arrange the transaction. Given the relevance of the choice for the household budget, it is likely that borrowers collect all relevant information before making a decision. With additional information on advertising, analyzing its effect for mortgage choice using a limited-information model represents an interesting area for future research.

rate r_{jt} and maturity T_i is given by

$$PV(q_{ijmt}, r_{jt}, c_{jt}, T_i) = q_{ijmt} \sum_{t=1}^{T_i} \left[\frac{r_{jt}(1+r_{jt})^{T_i}}{(1+r_{jt})^{T_i} - 1} - \frac{c_{jt}(1+c_{jt})^{T_i}}{(1+c_{jt})^{T_i} - 1} \right], \quad (4)$$

where q_{ijmt} is the outstanding mortgage quantity and c_{jt} is the marginal cost of providing mortgage j in quarter t .

Equation (4) does not account for the two key risks in the mortgage market that I discuss in Section II, namely, default and refinancing. First, default risk raises the lender's expected cost of issuing a mortgage. I assume that lenders setting interest rates do not forecast the probability of default in each period, but consider an average expected probability of default as in Crawford, Pavanini, and Schivardi (2018). Second, given the high level of refinancing at the end of the initial period, it is unreasonable to assume that lenders compute the present value as if all mortgages are held until maturity. I assume that lenders expect borrowers to refinance at the end of the initial period.³⁴

For long maturity, I can extend the approximation of equation (4) to include refinancing risk and default risk as follows:

$$\begin{aligned} PV(q_{ijmt}, r_{jt}, c_{jt}, t_j, d_{ijmt}) &\approx q_{ijmt} t_j (r_{jt} - c_{jt}) - d_{ijmt} q_{ijmt} t_j r_{jt} \\ &= q_{ijmt} t_j r_{jt} (1 - d_{ijmt}) - q_{ijmt} t_j c_{jt}, \end{aligned} \quad (5)$$

where t_j is the initial period in which the interest rate r_{jt} is fixed and d_{ijmt} is the expected default probability of borrower i with product j in market m and quarter t , which I also allow to be a function of the interest rate.³⁵

In each quarter t , lenders decide the initial rate for each product j they offer, taking as given the rates set by their competitors. Given the demand system and the approximation of the present value of net revenue from interest

³⁴ In Section III of the Internet Appendix, I derive a more flexible specification in which some borrowers fail to refinance at the end of the initial period and pay the SVR until maturity. As discussed in Section I.A, the majority of borrowers refinance their mortgage at the time their initial rate expires; hence, the initial rate is likely to be the major source of revenues for lenders. However, failure to refinance combined with high reversion SVR may also represent a profitable source of revenues for lenders (Belgibayeva et al., 2020). In the period I analyze, there is almost no variation within lender over time and across products in the SVR, despite the large variation in risk weights and initial rates. Controlling for lender and quarter fixed effects, the interquartile range of residuals is about 100 basis points for initial rate, while less than 1 basis point for SVR. Thus, the key mechanism in the paper based on differential regulation is less important for SVR, and differences across lenders are captured by the lender dummies when I take the model to the data.

³⁵ Equation (5) assumes that the remaining interest payment is lost upon default. In the case of collateralized lending, such as mortgage lending, the lender may be able to recover some fraction of the balance from the house sale and further actions against the borrower in default. Adding a positive recovery rate in the case of default would increase complexity of the model, requiring that lenders form expectations on future house price values and on the cost of collecting debt, which I do not observe in the data.

payments (5), I can write the lender's problem as

$$\begin{aligned}
 \max_{r_{jt}} \Pi_{lt}(r_{jt}) &= \sum_{j \in J_{lt}} \Pi_{jt}(r_{jt}) \\
 &= \sum_{j \in J_{lt}} \sum_m \sum_{i \in I_{mt}} s_{ijmt}(r_{jt}, r_{-jt}) \times PV(r_{jt}) \\
 &= \sum_{j \in J_{lt}} \sum_m \sum_{i \in I_{mt}} s_{ijmt}(r_{jt}, r_{-jt}) \times q_{ijmt}(r_{jt}) \times [t_j r_{jt}(1 - d_{ijmt}(r_{jt})) - t_j c_{jt}],
 \end{aligned} \tag{6}$$

where s_{ijmt} is the unconditional probability that borrower i chooses product j in market m in quarter t . I sum over all products j offered by lender l in quarter t ($\sum_{j \in J_{lt}}$), across all markets ($\sum_m$), and across all borrowers in market m and quarter t ($\sum_{i \in I_{mt}}$) to compute expected demand at the lender level. Note that the price of other products (r_{-jt}) enters the product demand (s_{ijmt}), but not the present value, which depends only on the conditional loan demand (q_{ijmt}). The derivative of the profits with respect to the price of product j is given by

$$\begin{aligned}
 \frac{\partial \Pi_{jt}}{\partial r_{jt}} &= \sum_m \sum_i s_{ijmt} q_{ijmt} (1 - d_{ijmt}) t_j + \sum_m \sum_i s_{ijmt} \frac{\partial q_{ijmt}}{\partial r_{jt}} [t_j r_{jt}(1 - d_{ijmt}) - t_j c_{jt}] \\
 &\quad + \sum_{k \in J_l} \sum_m \sum_i \frac{\partial s_{ikmt}}{\partial r_{jt}} PV_{ikmt} - \sum_m \sum_i s_{ijmt} q_{ijmt} \frac{\partial d_{ijmt}}{\partial r_{jt}} (t_j r_{jt}) = 0.
 \end{aligned} \tag{7}$$

The first term gives the extra profits from the higher rate on the quantity sold, the second term captures the change in loan demand from the higher rate, the third term gives the effect of the higher rate on the choice probability for all products offered by the lender, and the last term captures the effect of the higher rate on the default probability. Solving for the optimal interest rate gives

$$\begin{aligned}
 r_{jt}^* &= \frac{\overbrace{\sum_m \sum_i \left((1 - d_{ijmt}) - \frac{\frac{\partial d_{ijmt}}{\partial r_{jt}}}{\frac{\partial s_{ijmt}}{\partial r_{jt}} \frac{1}{s_{ijmt}} + \frac{\partial q_{ijmt}}{\partial r_{jt}} \frac{1}{q_{ijmt}}} \right)}^{\text{Effective marginal cost}}}{\underbrace{\sum_m \sum_i \left(\frac{\partial s_{ijmt}}{\partial r_{jt}} \frac{1}{s_{ijmt}} + \frac{\partial q_{ijmt}}{\partial r_{jt}} \frac{1}{q_{ijmt}} - \frac{\partial d_{ijmt}}{\partial r_{jt}} \frac{1}{1 - d_{ijmt}} \right)}_{\text{Full markup}}} \\
 &\quad - \frac{1}{\sum_m \sum_i \left(\frac{\partial s_{ijmt}}{\partial r_{jt}} \frac{1}{s_{ijmt}} + \frac{\partial q_{ijmt}}{\partial r_{jt}} \frac{1}{q_{ijmt}} - \frac{\partial d_{ijmt}}{\partial r_{jt}} \frac{1}{1 - d_{ijmt}} \right)} \\
 &\quad - \frac{\sum_m \sum_i \frac{\partial s_{ikmt}}{\partial r_{jt}} PV_{ikmt}}{\underbrace{\sum_{k \neq j \in J_l} t_j \sum_m \sum_i \left(\frac{\partial s_{ijmt}}{\partial r_{jt}} q_{ijmt} (1 - d_{ijmt}) + s_{ijmt} \frac{\partial q_{ijmt}}{\partial r_{jt}} (1 - d_{ijmt}) - s_{ijmt} q_{ijmt} \frac{\partial d_{ijmt}}{\partial r_{jt}} \right)}_{\text{Other products}}}.
 \end{aligned} \tag{8}$$

Note that if there is no default risk ($\frac{\partial d_{ijmt}}{\partial r_{jt}} = 0$ and $d_{ijmt} = 0$), all lenders offer only one product, and all households make only the discrete product choice ($q_{ijmt} = 1$), then equation (8) collapses to the standard markup pricing formula:

$$r_{jt}^* = c_{jt} - \frac{1}{\sum_m \sum_i \frac{\partial s_{ijmt}}{\partial r_{jt}} \frac{1}{s_{ijmt}}}.$$

Equation (8) characterizes the unconstrained optimal interest rate for lenders, but in reality, lenders set rates accounting for regulatory constraints in place. In the model, I focus on the two leverage regulations for the United Kingdom that I discuss in Section II.A. First, I add a risk-weighted capital constraint to the bank optimization problem. Even if lenders' balance sheets have assets other than mortgages, I assume that when they set rates for mortgages, they account for the capital requirement constraint. Second, I embed regulation on household leverage in the model by imposing a quota on the share of mortgages with an LTI above 4.5.

The problem for constrained lenders becomes

$$\begin{aligned} \max_{r_{jt}} \Pi_{lt}(r_{jt}) &= \sum_{j \in J_{lt}} \Pi_{jt}(r_{jt}) \\ \text{s.t. } \underline{K}_{lt} \sum_{j \in J_{lt}} \sum_m \sum_{i \in I_{mt}} s_{ijmt} q_{ijmt} \rho_{jt} &\leq K_{lt} && \text{Capital constraint} \\ \frac{\sum_{j \in J_{lt}} \sum_m \sum_{i \in I_{mt}} s_{ijmt} \mathbb{I}[LTI > 4.5]}{\sum_{j \in J_{lt}} \sum_m \sum_{i \in I_{mt}} s_{ijmt}} &\leq \bar{S} && LTI \text{ constraint}, \end{aligned}$$

where K_{lt} denotes the actual capital resources of lender l in quarter t , $\underline{K}_{lt}$ is the lender-specific minimum capital requirement in quarter t , ρ_{jt} is the risk-weight for mortgage product j in quarter t , $\mathbb{I}[LTI > 4.5]$ is an indicator for mortgages with an LTI greater than 4.5, and $\bar{S}$ is the maximum fraction of mortgages above the regulatory limit that lenders can issue in quarter t (e.g., 15% (Bank of England (2014))). The Lagrangian multipliers associated with the constraints represent the shadow cost of leverage regulations.

For example, adding the capital constraint and solving for the optimal rate for product j in quarter t gives

$$\begin{aligned} r_{jt}^* &= \frac{\overbrace{c_{jt} + \psi \underline{K}_{lt} \rho_{jt}}^{\text{Effective marginal cost}}}{\sum_m \sum_i \left((1 - d_{ijmt}) - \frac{\frac{\partial d_{ijmt}}{\partial r_{jt}}}{\frac{\partial s_{ijmt}}{\partial r_{jt}} \frac{1}{s_{ijmt}} + \frac{\partial q_{ijmt}}{\partial r_{jt}} \frac{1}{q_{ijmt}}} \right)} \\ &\quad - \frac{\overbrace{1}^{\text{Full markup}}}{\sum_m \sum_i \left(\frac{\partial s_{ijmt}}{\partial r_{jt}} \frac{1}{s_{ijmt}} + \frac{\partial q_{ijmt}}{\partial r_{jt}} \frac{1}{q_{ijmt}} - \frac{\partial d_{ijmt}}{\partial r_{jt}} \frac{1}{1 - d_{ijmt}} \right)} \end{aligned}$$

$$- \sum_{k \neq j \in J_l} \frac{\sum_m \sum_i \frac{\partial s_{ikmt}}{\partial r_{jt}} PV_{ikmt}}{t_j \sum_m \sum_i \underbrace{\left(\frac{\partial s_{ijmt}}{\partial r_{jt}} q_{ijmt} (1 - d_{ijmt}) + s_{ijmt} \frac{\partial q_{ijmt}}{\partial r_{jt}} (1 - d_{ijmt}) - s_{ijmt} q_{ijmt} \frac{\partial d_{ijmt}}{\partial r_{jt}} \right)}_{\text{Other products}}}, \quad (9)$$

where ψ is the Lagrangian multiplier for the capital constraint, which measures the extent of the cost when a bank's capital is close to the risk-adjusted requirement.³⁶ Hence, the marginal cost now accounts for both the direct cost of originating the mortgage c_{jt} and the opportunity cost due to its impact on the capital constraint $\psi \underline{K}_{jt} \rho_{jt}$.

The equilibrium in the mortgage market is characterized by both lenders' optimal pricing subject to leverage regulations and borrowers' optimal mortgage choice taking as given prices and other product characteristics.³⁷

IV. Estimation and Identification

In this section, I discuss how I take the model to the data. First, I describe how I build borrowers' choice sets, which are not observable in the data and could be affected by affordability criteria. Next, I discuss the estimation approach and the variation that I use for identification, endogeneity concerns, and supply-side instruments.

A. Counterfactual Choice Set

I determine the products available in borrowers' choice set in three steps. First, I break the population into markets defined by four demographic groups based on below- and above-median age and income jointly with the region in which the borrower's house is located. I assume that borrowers in each market can access all products sold in that market in the same quarter in which they borrow, but not those sold in other markets or other quarters. This restriction comes from the fact that borrowers with similar observable characteristics have access to similar alternatives. Restricting the choice set using some borrowers' demographics could also partially compensate for the absence of data on mortgage applications and on banks' loan approval decisions.³⁸

³⁶ If the capital requirement is not binding, then ψ would be equal to zero in the model. In reality, banks target a buffer over the minimum requirements, and thus, in the estimation I allow banks with different levels of capital buffer to have different costs. In Section III of the [Internet Appendix](#), I discuss an extension of the model to the case in which lenders can also choose risk weights.

³⁷ While the assumptions used to derive equation (9) are tailored to the U.K. mortgage market, the model is quite flexible and can be applied to other contexts. For example, if credit scores affect mortgage rates as in the United States and are observable to the econometrician, they can be added as an additional characteristic. Including more observable product or borrower characteristics affecting mortgage rates will increase the number of first-order conditions without changing to the main framework.

³⁸ For example, a credit card company reports that in the United Kingdom, rejection rates in the mortgage market are low but vary with demographics such as age (observable), which can be cor-

Given the national nature of the market, I do not impose restrictions on the choice set based on locations finer than the level of the region. My analysis focuses on the largest lenders, with portfolios of mortgages from across the United Kingdom, and even products from smaller lenders, with a more local business model, can be sold nationally via Internet, the phone, and brokers. Ruling out products from households' choice sets based on their within-region location seems to be somewhat extreme and unrealistic in a market such as the United Kingdom. However, I allow for a narrower geographical level to play a role through the local branch network of the lender.

Second, in each market-quarter, I focus on the most popular mortgages offered by the 13 largest lenders, which account for more than 90% of mortgage originations in the period I analyze. The U.K. mortgage market has historically been very innovative, with a large number of products offered (Cocco (2013)). However, a lot of these products have very small market shares. For example, while mortgages are available with an initial fixed-rate period of 1 to 10 years, mortgages with a two-year fixed period account for more than 60% of originations in my sample. Accordingly, I group mortgages offered by lenders outside the 13 largest lenders and mortgages with a market share below 0.3% in a representative "outside" product.³⁹

Finally, I implement an additional step within market-quarter to further restrict borrowers' choice set. A drawback of the approach to construct the choice set so far is that I can include products that are not in borrowers' choice sets, which can lead to biased parameter estimates (Goeree (2008), Gaynor, Propper, and Seiler (2016)). For example, maximum LTI and LTV limits can impose an upper bound on borrowers' leverage, while unobservable differences in wealth can act as a lower bound. In the data, I observe borrowers in the same market and quarter shopping at very different maximum LTV bands. Thus, I further restrict borrower i 's choice set to only those mortgages with a maximum LTV equal to the one chosen by i , which I observe, and just above and below, which I observe for borrowers in the same market and quarter of borrower i .⁴⁰ Thus,

related with credit history (unobservable to the econometrician but observable to the lender); see <https://www.creditladder.co.uk/blog/mortgage-applications-rejected>. In a recent paper, Crawford, Griffith, and Iaria (2016) describe the use of the choice set of similar consumers as the interpersonal logit model.

³⁹ Among others, Goeree (2008) studies households' choice of personal computer and consider as the outside good nonpurchase, purchases of used computers, and purchases of new computers from firms not in the data. In applications to financial markets, Egan, Hortaçsu, and Matvos (2017) study households' choice of deposits and consider as the outside good banks outside the top 16.

⁴⁰ As an example, in equilibrium, a borrower buying a product with a maximum LTV of 90% will have in the choice set mortgages with a maximum LTV of 90%, 85%, and 95%. I also estimate the model without this additional restriction and find that borrowers have positive, even if not significant, demand elasticities. The reason for this result in the larger choice set is perhaps that I am including in borrowers' *counterfactual* choice set (low rate) products that are not in borrowers' *true* choice set. Indeed, Figure 5 shows that a lot of bunching takes place at high-LTV bands, such as 90% and 95%. The fact that these high-leverage borrowers do not choose low LTV, low-rate products when included in their choice set, generates a low elasticity to rates, all else equal. The potential sensitivity of demand estimates to different assumptions about products in individuals'

I allow borrowers to shop locally in terms of the downpayment decision, consistent with the bunching behavior in Figure 5.

B. Estimation

B.1. Demand

My demand model in Section III.A predicts households' mortgage demand and loan size as a function of observable household characteristics, random preferences, product attributes, and a vector of parameters to be estimated. Building on the discrete-continuous choice literature (Dubin and McFadden (1984), Train (1986)), I make the following parametric assumption on the indirect utility $\bar{V}_{ijmt}$:

$$\bar{V}_{ijmt} = \frac{\gamma_m}{1 - \phi_m} Y_i^{1 - \phi_m} + \mu_m \exp(\overbrace{-\alpha_m r_{jt} + \beta_m X_j + \xi_{jmt}}^{\delta_{jmt}} + \eta_m D_i + \zeta_i) + \lambda_m B_{ij(l)}, \quad (10)$$

where all demand parameters are allowed to vary across markets m and δ_{jmt} captures all observed and unobserved product attributes. I also make two common assumptions on the structural unobservable error terms. First, I assume that ε_{ijmt} in equation (1) is identically and independently distributed across households and mortgage products with a type I extreme value distribution. Second, I assume that borrowers' unobservable heterogeneity follows a normal distribution with mean zero and variance σ_m .

The conditional probability that borrower i chooses product j in market m and quarter t is then given by

$$p_{ijmt}(\zeta_i) = \frac{\exp(\bar{V}_{ijmt})}{\sum_{j'=0}^{J_i} \exp(\bar{V}_{ij'mt})}, \quad (11)$$

and the unconditional probability that borrower i chooses product j in market m and quarter t equals

$$s_{ijmt} = \int_{\zeta} p_{ijmt}(\zeta_i) dF(\zeta_i). \quad (12)$$

Using Roy's identity, I obtain the loan demand function q_{ijmt} conditional on choosing mortgage j :

$$\ln(q_{ijmt}) = \ln \left(-\frac{\frac{\partial \bar{V}_{ijmt}}{\partial r_{jmt}}}{\frac{\partial \bar{V}_{ijmt}}{\partial Y_i}} \right) = \ln \left(\frac{\mu_m \alpha_m}{\gamma_m} \right) + \phi_m \ln(Y_i) + \delta_{jmt} + \eta_m D_i + \zeta_i. \quad (13)$$

choice sets is in line with recent papers that take an in-depth look at the formation of choice sets (Abrams (2020), Abaluck and Adams-Prassl (2021)).

From equation (13) and the normal distribution assumption for ζ_i , the probability of the conditional loan demand is

$$f(\ln(q_{ijmt})|j, j \neq 0) = \frac{1}{\sqrt{2\pi\sigma_m^2}} \times \exp \left[-\frac{\left(\ln(q_{ijmt}) - \left(\ln\left(\frac{\mu_m \alpha_m}{\gamma_m}\right) + \phi_m \ln(Y_i) + \delta_{jmt} + \eta_m D_i \right) \right)^2}{2\sigma_m^2} \right]. \quad (14)$$

Therefore, the joint log likelihood of individual i buying product j and borrowing amount q_{ijmt} in market m and quarter t is given by

$$\ln(L_i) = \sum_{j=0}^{J_i} \mathbb{I}_{ijmt} [\ln(s_{ijmt}) + \ln(f(\ln(q_{ijmt})|j, j \neq 0))], \quad (15)$$

where $\mathbb{I}_{ijmt}$ is an indicator variable equal to 1 if borrower i chooses product j and 0 otherwise.

I estimate the demand model in two steps. In the first step, I estimate simultaneously the discrete-continuous choice by maximizing the joint log-likelihood function for each market summing across all borrowers and quarters, that is, $\mathcal{L}_m = \sum_t \sum_i \ln(L_i)$. The joint likelihood addresses the simultaneity bias that arises if I do not account for the discrete product choice (equation (12)) when I estimate the continuous quantity choice (equation (13)). I also account for possible correlation between the interest rate r_{jmt} and unobservable product characteristics ξ_{jmt} by including a full set of product-market-quarter fixed effects δ_{jmt} in the log-likelihood function. Thus, in the first step, I obtain the utility parameters $\phi_m, \eta_m, \lambda_m$, the scaling factors σ_m and μ_m , and the product-market-quarter fixed effects δ_{jmt} .

In the second step, I recover the impact of the interest rate r_{jt} and of other product characteristics X_j using an instrumental variable strategy whose dependent variable is the product-market-quarter fixed effect δ_{jmt} estimated in the first step. Specifically, I estimate the regression

$$\hat{\delta}_{jmt} = -\alpha_m r_{jt} + \beta_m X_j + \gamma_t + \xi_{jmt}. \quad (16)$$

In Section IV.C, I discuss the supply-side instruments that I use to obtain consistent estimates of the parameters in equation (16).

B.2. Supply

The estimation of the supply-side parameters relies on the optimal pricing formula derived in Section III.B. Specifically, I back out the unobservable marginal costs at the product-quarter level by inverting equation (9) and

solving for $mc_{jt} = c_{jt} + \psi \underline{K}_{lt} \rho_{jt}$. Marginal costs are a function of observed interest rates and the estimated demand elasticities. To compute the full markup given by the second term of equation (9), I also need information on the average expected default rate and on the increase in defaults as a result of changes in interest rates. I back out the latter using a cross-section of data on mortgage performances as of June 2016 and the linear probability model given by

$$d_{ijmt} = \iota_r^d r_{jt} + \iota_X^d X_j + \iota_D^d D_i + \iota_t^d + \varrho_{ijmt}^d, \quad (17)$$

where d_{ijmt} is a dummy equal to 1 if borrower i who took product j in market m and quarter t is in arrears in June 2016, r_{jt} is the interest rate, X_j is a vector of product j characteristics at origination (e.g., LTV), D_i is a vector of borrower i characteristics at origination (e.g., age, income), and ι_t^d are cohort fixed effects.

It is important to note that equation (17) estimates the effect of variables at origination on ex-post outcomes. As some variables will change over time (e.g., income, house value), a fully specified model should control for the actual value of these variables. Unfortunately, I do not have a panel that will allow me to do that. However, variables at origination play an important role in pricing the *expected* probability of arrears, which is the object of interest in the pricing function.⁴¹

With the demand and default estimates, as well as the observed interest rates, I have the elements from equation (9) needed to back out unobserved marginal costs. I then let marginal cost be a flexible function of observed product characteristics, observed cost shifters, and other unobservable characteristics and determinants of costs,

$$mc_{jt} = \tau_Z^c Z_{jt} + \tau_X^c X_j + \tau_t^c + \kappa_{jt}^c, \quad (18)$$

where Z_{jt} are observed cost shifters (risk-weighted capital requirements, funding spread, and swap rates), X_j are observed product characteristics (interest rate type, maximum LTV, and lender fixed effects), τ_t^c are quarter fixed effects, and κ_{jt}^c is a structural error term capturing unobservable determinants of marginal costs (e.g., advertising expenditures).

C. Identification

C.1. Demand

Estimation of the demand parameters faces two endogeneity concerns arising from the simultaneity problem and from unobservable attributes affecting

⁴¹ I use the estimated parameters from equation (17) in my counterfactual exercises, as changes in costs impact arrears. Specifically, lenders pass changes in costs onto interest rates, which have a direct impact on arrears, as captured by ι_r^d in equation (17), and an indirect impact via both the discrete mortgage choice and the continuous quantity choice. To use the parameters from the default model in the counterfactual analysis, I need to assume that the change in regulation does not change the relation between the interest rate and default. In Section VI, I test the robustness of my results using external estimates for the parameter ι_r^d .

demand. First, I address the simultaneity problem by estimating the discrete-continuous choice jointly, as shown in equation (15). This approach allows me to address the bias that can arise if I do not account for the discrete product choice when I estimate the continuous quantity choice. To separately identify the discrete and continuous choices, I assume that lenders' local branch presence affects borrowers' choice among lenders, but not the conditional choice of quantity. Since I estimate the model in each region separately and I control for lender fixed effects, my assumption requires that within a region, a larger branch presence in a postcode area does not differentially affect the loan demand of borrowers choosing that lender.⁴² I exploit variation in the branch network together with variation in the location of the borrowers' houses at the postcode level to identify the effect of branches on product demand.

Second, lenders endogenously choose interest rates, which could be correlated with unobservable attributes that also affect demand (e.g., advertising, screening, cash back). As an example, consider a lender that relaxes screening or offers cash back for a specific product and at the same time increases the interest rate on that product. Screening efforts and the cash-back option are not observable to the econometrician, and thus enter the error term (ξ_{jmt}), and at the same time are correlated with the interest rate r_{jmt} . As a result, I can observe borrowers choosing the product and mistakenly conclude that they are not responding to the higher interest rate, while the effect of the higher rate is countervailed by the differential screening effort or the cash-back option.

To account for endogeneity in the interest rate, I adopt a two-step estimation approach. The first step includes product-market-time fixed effects δ_{jmt} that capture all of the variation in mortgage rates. The second step from regression (16) addresses correlation between mortgage rates r_{jmt} and unobservable product characteristics ξ_{jmt} using an instrumental variable strategy. Most notably, I exploit the institutional features of the leverage regulation in place in the United Kingdom that I described in Section II.A to construct cost shifters at the *product-time* level. I instrument interest rates using variation in risk weights that affects the cost to lenders of issuing a particular product. The identification assumption for the demand parameters is

$$E[\xi_{jmt}|X_j, Z_{jt}, \gamma_t] = 0, \quad (19)$$

where X_j are product-level controls, γ_t are quarter fixed effects, and Z_{jt} are my instruments. As product characteristics I include a dummy for mortgages with high LTV (above 85%), a dummy for mortgages with a longer fixed period (5 years), and lenders fixed effects. The lender and quarter fixed effects control nonparametrically for time-invariant average unobservable differences across lenders and for time-varying common factors that affect mortgage rates.

⁴² This exclusion restriction is supported by the empirical evidence. In Section I of the [Internet Appendix](#), I regress market shares and loan amounts on quartiles of branches, controlling for differences across lenders and postcodes with fixed effects. Figure IA.3 shows that a larger branch presence affects lenders' market share, but has no differential effect on loan amounts. Furthermore, my assumption is in line with previous studies estimating discrete-continuous supermarket choice models (Smith (2004), Dubois and Jódar-Rosell (2010)).

The endogenous interest rate r_{jt} is instrumented in the baseline specification with the lender risk weights ρ_{jt} , and I also estimate a specification with lender risk-weighted capital requirements $\underline{K}_{lt}\rho_{jt}$. The minimum capital requirements $\underline{K}_{lt}$ vary across lenders and over time at the quarterly frequency; while the risk weights ρ_{jt} vary across lenders, across LTV bands, and over time at the yearly frequency. Thus, the identifying variation for the endogenous interest rate is along three dimensions: (i) controlling for LTV and time, across lender because of different risk weights and minimum capital requirements; (ii) controlling for lender and time, across LTV bands because of different risk weights; and (iii) controlling for lender and LTV, over time because of changes in risk weights for lenders with IRB model and changes in minimum capital requirements for all lenders.

Equation (19) indicates that predetermined variation in risk weights (risk-weighted capital requirements) is uncorrelated with demand shocks, conditional on a rich set of macro-, lender-, and product-level controls. The intuition for this identifying assumption is that mortgage borrowers care about risk-weighted capital requirements only insofar as they affect mortgage rates. Endogeneity in the regulatory instrument Z_{jt} that is correlated with unobservable household preferences ξ_{jmt} can pose a threat to my identification strategy. Institutional features, additional evidence, and robustness tests mitigate this concern.

First, my identification assumption is conditional on mortgage product characteristics, which include lender fixed effects and controls for high-LTV mortgages, thus requiring a *differential* change in risk weights across LTVs *within* lender, as an average change will be captured by the fixed effects. Furthermore, changes in the internal models need to be approved by the regulators, limiting lenders' discretion in setting them.⁴³ Second, the reduced-form evidence in Section II.D shows that lenders using internal models do not appear to be different from lenders adopting the standardized approach across LTV bands in both ex-ante borrower characteristics (age and income) and ex-post performance (arrears and refinancing). Third, I test the robustness of my demand estimates by instrumenting endogenous mortgage rates with the risk weights of *other* lenders.

While the key endogenous variables in my setting are mortgage rates, I briefly discuss the variation that allows me to identify other demand parameters. The covariation between the exogenous product characteristics X_j and borrowers' choices identifies the other parameters β_m in equation (16). The product-market-time fixed effects δ_{jmt} are identified in the first step from variation in market shares across products and over time, and correlation between

⁴³ In a recent paper, Behn, Haselmann, and Vig (2016b) show that lenders with IRB models underreport risk weights. Differences in the decision to adopt IRB models are not a concern for my identification strategy because: (i) the vast majority of lenders switched to IRB in 2008 and (ii) lender fixed effects control nonparametrically for lender-wide differences in the benefits of adoption and reported risk weights. Thus, only differential reporting within lender across LTVs could be a concern for the validity of the instrument to the extent that this behavior is also correlated with unobservable factors affecting borrowers' utility.

the loan amount and the product choice in a market-quarter.⁴⁴ The scaling factor μ_m is identified by the different weight of the product-market-time fixed effects δ_{jmt} on the discrete relative to the continuous choice, while the scaling factor σ_m is identified by individual-level variation in both product and quantity choice within market that is not explained by other demographics. Correlation between the continuous loan amount and income within product-market-time across borrowers identifies the elasticity of loan demand to income ϕ_m . Finally, the parameter η_m , which captures the effects of demographics such as age and local house prices, is identified by both variation in loan demand within product-market-time and the correlation between borrowers' product choice and the interaction of product characteristics with borrowers' demographics.

C.2. Supply

Identification of the supply-side parameters τ_Z^c and τ_X^c in equation (18) comes from correlation between cost shifters Z_{jt} and product characteristics X_j with marginal costs mc_{jt} at the product level. The main cost shifters of interest for my analysis are risk-weighted capital requirements. As discussed in Section II.A, capital requirements vary both across lenders and over time, due to regulatory changes, and across products offered by the same lender, due to the risk-weights adjustment. Most notably, minimum capital requirements $\underline{K}_{lt}$ vary across lenders and over time at the quarterly frequency, while risk weights ρ_{jt} vary across lenders, across LTV bands, and over time at the yearly frequency. Therefore, I identify the shadow cost of risk-weighted capital regulation controlling for (i) differences across lenders that are common among products (lender shocks) and (ii) differences across products that are common across lenders (market shocks).

I include two additional variables in Z_{jt} that can affect marginal costs. First, a measure of the cost of lenders' wholesale funding based on secondary-market senior unsecured bond spreads. This measure is constructed by the Bank of England to track developments in lenders' funding cost and vary over time and across products based on the lender originating the mortgage. Second, I add the euro interest rates swaps for two and five years. Swap rates are a hedging instrument that lenders use when selling mortgages with fixed periods of two and five years, and thus, they vary over time and across products based on the initial fixed period.

I also allow the effect of risk-weighted capital requirements to vary across lenders. Heterogeneity in the effect of risk-weighted capital requirements is

⁴⁴ I reestimate my model shutting down the continuous choice to explore the role of the quantity choice and the cross-equation restrictions in identifying different parameters. As expected, the correlation in the point estimates between the full model and the discrete-only choice model is higher for parameters such as lender fixed effects in X_j , for which variation in loan demand is less important for identification, than for parameters like α_m , for which the continuous choice is more important for identification.

consistent with both my model and previous literature (Gambacorta and Mishtrulli (2004), Kashyap, Stein, and Hanson (2010), Aiyar et al. (2014), Jiménez et al. (2017)).⁴⁵ Hence, I estimate equation (18) with an additional term obtained interacting risk-weighted capital requirements with a dummy for lenders with a capital buffer (wholesale funding cost) above the median.

I perform two tests to check the robustness of my estimates of the effect of risk-weighted capital requirements on product-level marginal costs. First, I estimate a version of equation (18) with interacted lender-quarter fixed effects. This specification allows me to identify the effect of risk-weighted capital requirements using variation within lender-quarter jointly across products with different LTV bands and over time due to changes in risk weights. This robustness check highlights the advantage of modeling lenders as multiproduct firms, as it allows me to control for unobservable confounding factors that vary jointly over time and across lenders.⁴⁶

Second, to reduce concerns about endogeneity in risk weights for lenders using IRB models that could be correlated with omitted cost determinants κ_{jt}^c , I also estimate equation (18) using an instrumental variable strategy. Specifically, I exploit exogenous changes in capital requirements $\bar{K}_{lt}$ that take place at the end of 2015 for some lenders and compare the evolution of their marginal costs to other lenders who did not experience changes in capital requirements at the same time.

Finally, similar to the demand side, I also estimate the impact of exogenous product characteristics on marginal costs captured by the parameter τ_X^c in equation (18). The identification relies on how characteristics such as the length of the initial period, which captures interest rate risk, and maximum LTV, which captures default risk, covary with marginal costs.

V. Estimation Results

A. Demand Parameters

In this section, I present results from estimation of the structural model using data for first-time buyers. Table II shows the estimated demand parameters averaged across all markets.⁴⁷ The main parameter of interest is α , which

⁴⁵ According to the optimal pricing equation (9), if the risk-weighted capital constraint is not binding, its shadow cost should be zero. In practice, lenders target a buffer over the minimum requirements, so that the difference between actual resources over requirements, that is, the capital buffer, could affect the impact of changes in capital requirements on marginal costs. I expect lenders with higher buffers to be less sensitive to differences in risk-weighted capital requirements, as they have more margins to adjust relative to lenders with little or no extra capital. Furthermore, lenders with relatively high funding costs may be more sensitive to differences in risk-weighted capital requirements, as their costs of raising external funds are higher.

⁴⁶ Previous studies model the choice of lender and control for *average* differences across lenders with lender dummies but cannot control for unobservable *time-varying* differences across lenders (see, among others, Egan, Hortaçsu, and Matvos (2017) and Crawford, Pavanini, and Schivardi (2018)).

⁴⁷ Consistent with the reduced-form evidence discussed in Section II, leverage regulation has a larger impact on interest rates and credit access for borrowers with higher leverage, while home

Table II
Structural Demand Estimates

The table reports the structural demand estimates for the econometric demand model of Section IV.B. The model is estimated separately in each market and the table reports the average point estimate and standard error in each group. The total number of borrowers is 370,575 and the average number of product-market observations is 773. The standard errors for the parameters in the first stage are computed using the inverse of the information matrix; the standard errors for the mortgage attributes estimated in the second stage are computed by bootstrapping. *, **, and *** indicate significance at the 0.10, 0.05, and 0.01 levels, respectively.

	Structural Demand Parameters					
	Interest (α)	Leverage (β_1)	Fix Period (β_2)	Branches (λ)	Income (ϕ)	Heterogeneity ($\log(\sigma)$)
Average	-0.025*** 0.002	0.010*** 0.0019	0.025*** 0.003	0.019* 0.011	0.700*** 0.001	-1.608*** 0.009

captures the effect of the interest rate on indirect utility. As expected, I find that borrowers dislike more expensive mortgages and that the effect of mortgage rates on demand is highly significant. The point estimates across markets is approximately 0.025.

To understand the magnitude, I compute the discrete and continuous elasticities using the formulas reported in Section III of the Internet Appendix. I find an average loan demand elasticity of about 0.07 and a product demand elasticity of 5.9 across all mortgage products. The loan demand elasticity implies that a 1-percentage-point increase in the interest rate leads to a 2.5% change in loan demand. This number is in line with the estimates from previous studies using very different approaches, such as bunching techniques (DeFusco and Paciorek (2017)) and survey data (Fuster and Zafar (2021)).⁴⁸

The product demand elasticity implies that, on average, a 1% increase in the interest rate decreases the market share of the mortgage by almost 6%. When I focus on the 10 most popular products, I find that a 1% increase in the interest rate decreases the market share of the mortgage by 3% to 7%, while the shares of other mortgages increase by 0.01% to 0.07%. These own- and cross-price elasticities are in line with results in recent structural work looking at the U.K. and the U.S. mortgage markets (Buchak et al. (2018a),

movers and remortgagers may already have accumulated equity in their houses. Section IV of the Internet Appendix presents averages by income, age, and selected regions. I also report the *F*-statistic for the excluded instrument, which is above 150 across all markets.

⁴⁸ For example, DeFusco and Paciorek (2017) study the U.S. mortgage market and find that a 1-percentage-point increase in the interest rate on a 30-year fixed-rate mortgage reduces first mortgage demand by between 2% and 3%. Similarly, Fuster and Zafar (2021) find that a change in the mortgage rate of 2 percentage points changes willingness to pay by about 5% on average. In my setting, a 2-percentage-point increase in mortgage rate would decrease optimal loan amount by approximately 5%.

Table III
Loan Demand and Own Product Demand Elasticities

The table shows the interest rate elasticities for a random subsample of first-time buyers. I show the loan demand and own-product demand elasticities. The elasticities are computed using the structural parameters from Table II and the formulas in Section III of the Internet Appendix. I report the average elasticities across all products and by three product characteristics: lender type, maximum LTV, and fixed period.

	Loan Demand		Product Demand	
	Mean	SD	Mean	SD
All	-0.073	0.022	-5.935	1.704
Lender type				
Big 6	-0.073	0.022	-5.963	1.737
Challengers	-0.076	0.022	-6.147	1.724
Building societies	-0.073	0.022	-5.709	1.572
Maximum LTV				
LTV ≤ 70	-0.058	0.012	-4.801	0.989
70 < LTV ≤ 85	-0.065	0.015	-5.295	1.163
LTV > 85	-0.096	0.018	-7.676	1.433
Fixed period				
Two years	-0.065	0.021	-5.284	1.638
Five years	-0.083	0.019	-6.694	1.446

Robles-Garcia (2019)) and the Italian corporate loan market (Crawford, Pavanini, and Schivardi (2018)).⁴⁹

Given that my product definition combines several elements of horizontal differentiation, I can compute elasticities at various levels. In Table III, I report the average loan demand and own-product demand elasticity across mortgage characteristics. The six largest lenders and building societies have similar loan demand elasticity, while challenger banks face higher demand elasticity. In terms of product demand, building societies have the lowest elasticity, followed by the six largest lenders. Challenger banks face the highest elasticities of mortgage demand. As a result, for the same percentage point increase in interest rate, challenger banks lose more customers and face a larger decrease in loan demand from customers who buy their products. I also explore heterogeneity across leverage levels. Both loan and product demand elasticities increase with leverage. Mortgages with a maximum LTV above 85% have, on average, a loan demand elasticity of 0.09 and a product demand elasticity of 7.5 relative to mortgages with leverage below 70%, whose elasticities are 0.06 and 4.8, respectively.

Table II also reports the estimated demand parameters capturing preferences for other product characteristics, such as maximum leverage and the length of the fixed period, which play a central role in mortgage choice (Campbell and Cocco (2003), Badarinza, Campbell, and Ramadorai (2017)). I find

⁴⁹ Robles-Garcia (2019) finds that own-product demand elasticity is between 3 and 5 and cross-product demand elasticity is around 0.02 for the U.K. mortgage market, while Buchak et al. (2018a) find price elasticities ranging from 2 to 4 for the U.S. mortgage market.

that first-time buyers value mortgages with high leverage, which allow for lower downpayments for credit-constrained borrowers. I also find that borrowers prefer longer fixed-rate periods, as such contracts minimize switching costs involved in refinancing as well as interest rate risk. The fraction of branches in the postcode in which households have their houses has a positive but only marginally significant impact on average.

Finally, while so far I focus on average effects, the estimated model allows for rich heterogeneity, both observable and unobservable. Table II reports the effect of individual income (ϕ_m) and unobservable heterogeneity (σ_m). I find a positive and significant elasticity of loan demand to income with a magnitude of 0.7. I also find significant unobservable heterogeneity across households, even within my narrowly defined market and after controlling for observable demographics within market.⁵⁰

A.1. Fit and Robustness

In Section IV of the Internet Appendix, I look at the ability of the model to predict key variables of interest on the demand side, namely, loan demand, LTI, LTV and product market shares. Table IA.VI shows key moments predicted by the model and the corresponding moments in the data. Overall, the model fits the data well, in terms of both mean and variance. The main limitation is that the model underpredicts the variance in LTV shares, as it is not able to capture well the extreme leverage levels that are sometimes observed in the data.⁵¹ Table IA.VI also reports the out-of-sample fit of the model. Specifically, after estimating the structural demand parameters using data for 2015 and 2016, I predict loan demand, LTI, market shares, and LTV for mortgages in the first two quarters of 2017. Out of the estimation sample, the model captures the data well.

In Section V of the Internet Appendix, I discuss several robustness checks. First, I instrument the endogenous interest rate for first-time buyers with both the risk-weighted capital requirements, thus exploiting additional variation in capital requirements over time and across lenders, and the risk weights for the same maximum LTV of other lenders. Second, I construct an initial annual percentage rate (APR) as a function of both the interest rate and the lender fee for a representative loan amount and use it as the price of the mortgage instead of the initial rate only. Third, I estimate the second step of the demand model (equation (16)) simultaneously with the supply side (equation (18)) using

⁵⁰ Figure IA.7 in the Internet Appendix shows the normalized elasticity of demand to interest rate and LTV band in a market in the baseline model and in an alternative model in which I shut down the individual heterogeneity within market. The model with individual heterogeneity displays larger variation within market, generating more flexible substitution patterns.

⁵¹ To improve the fit at the individual level, additional information on individuals' construction of the choice set would be needed (e.g., credit score, value of houses considered, and saving accumulated for downpayment). However, under the assumption on the choice set discussed in Section IV.A, the model performs well in predicting the choice of LTV band with a correlation of 0.9 between predicted and observed market shares.

Table IV
Default Estimates

The table reports the default estimates from equation (17). The dependent variable is a dummy equal to 1 if a mortgage originated between 2005 and 2015 is in arrears in June 2016. I define arrears as being at least three months late in servicing the monthly payment. Column (1) uses the full sample and column (2) focuses on mortgages originated before 2008, while columns (3) and (4) look at mortgages originated after 2008. Interest is the interest rate at origination expressed in percentage terms. High LTI is a dummy for mortgages with LTI at origination above 3.5. High LTV is a dummy for mortgages with LTV at origination above 85%. Borrower controls include type of borrower, employment status, income, age, maturity, and property value. In column (4), the excluded instrument for the interest rate is the mortgage risk weight and swap rate. Robust standard errors are in parentheses. *, **, and *** indicate significance at the 0.10, 0.05, and 0.01 levels, respectively.

	Full Sample	Precrisis	Postcrisis	
	OLS (1)	OLS (2)	OLS (3)	IV (4)
Interest (%)	0.0025*** (0.0003)	0.0080*** (0.0015)	0.0015*** (0.0002)	0.0021*** (0.0007)
High LTI	0.0012*** (0.0003)	0.0060*** (0.0015)	0.0002 (0.0002)	0.0003* (0.0002)
High LTV	-0.0015** (0.0006)	0.0158*** (0.0022)	-0.0015*** (0.0003)	-0.0020*** (0.0007)
Time F.e.	Yes	Yes	Yes	Yes
Lender F.e.	Yes	Yes	Yes	Yes
Rate type F.e.	Yes	Yes	Yes	Yes
Postcode district F.e.	Yes	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes
F stat				229
Arrears (mean)	0.01	0.06	0.01	0.01
Observations	1,636,074	218,707	1,417,295	1,394,177

generalized method of moments. The demand estimates are robust to these different instruments, variable definitions, and estimation methods.

B. Supply Parameters and the Cost of Capital Regulation

In this section, I present results from the estimation of the supply-side parameters. First, I discuss estimates of the default model given by equation (17), whose parameters enter the calculation of markups and effective marginal costs. Table IV reports the results. Column (1) uses the full cross-sectional data on arrears. I find a statistically significant positive relation between the interest rate and default. In terms of magnitude, my estimates imply that a 1-percentage-point higher interest rate increases the probability of default by 0.25 percentage points. Given an average default rate of about 1.3%, a 1-percentage-point increase in rate generates approximately a 20% increase in arrears.

I employ the estimates in column (1) in the baseline simulation, but I also explore alternative specifications for robustness. First, in columns (2) and (3) of Table IV, I split my sample into mortgages originated before and after the crisis. The relation between interest rate and default is positive and significant in both periods. The point estimate is higher before the crisis. For mortgages originated before the crisis, a 1-percentage-point higher interest rate increases the probability of default by 0.8 percentage points, while the effect is approximately five times smaller for mortgages issued after the crisis. Note, however, that average default is about 6% for mortgages originated before the crisis and only 1% for mortgages originated after the crisis. Thus, the effect of a 1-percentage-point higher interest rate relative to the mean level of arrears is approximately 14% in both sub-samples.

Second, in column (4) of Table IV, I address endogeneity concerns coming from omitted variables correlated with both the interest rate and the default probability. Specifically, I instrument the interest rate in equation (17) with supply-side cost shifters (i.e., risk weights and swap rates). I focus on mortgages originated after the crisis given that I do not have variation in risk-weights precrisis. The instrumental variable estimate of the marginal effect of mortgage rates on default is slightly stronger than the OLS estimate in column (3). A 1-percentage-point higher interest rate increases the probability of default by about 0.2 percentage points. The magnitude is in line with the point estimate using the full sample in column (1) of Table IV.

Third, the magnitudes that I find are in line with the estimates in previous studies that explore the effect of marginal changes in loan rate on firms' default (Crawford, Pavanini, and Schivardi (2018)), and use quasi-natural variation in mortgage rates to understand borrower default (Fuster and Willen (2017), Gupta (2019)). The most comparable estimates are by Fuster and Willen (2017), who find that a 2-percentage-point reduction in the interest rate decreases the probability of default by about 40%. In my setting, I find that a 2-percentage-point increase in rate leads to an increase in the probability of default of about 38%.⁵² While my estimates are in line with previous work, in the counterfactual analysis in Section VI, I also test the robustness of my results to doubling the effect of mortgage rates on default.

Finally, in Table IV, I also study how LTI and LTV at origination affect the probability of default. These two variables affect counterfactual default and mortgage rates as they endogenously respond in equilibrium to changes in mortgage rates via both the discrete product choice, which determines the LTV band, and the continuous loan choice, which for a given income determines the LTI. Column (1) shows that mortgages with LTI above 3.5 are significantly more likely to default, as expected, while mortgages with an LTV above 85% are less likely to default, which is perhaps surprising. The analysis by

⁵² The slightly smaller effect may be driven by asymmetric effects in response to increases relative to decreases. Also, Fuster and Willen (2017) note that the point estimates of the effects of rate reductions tend to be smallest, and thus more in line with my estimates, for borrowers who are not (or only mildly) underwater, which is the case in my setting with increasing house prices.

mortgages originated before and after the crisis helps explain this pattern. Controlling for the interest rate, high-LTV mortgages issued before the crisis are significantly more likely to default, while high-LTV mortgages issued after are significantly less likely to default. A possible explanation for the latter result may be the increase in supply-side restrictions and affordability checks after the crisis, which led to an overall low volume of originations at high leverage to a select pool of low-risk borrowers.

Using lenders' first-order conditions and the estimated demand and default parameters, I compute product-level markups and marginal costs. The average markup is about 0.50 percentage points in the full sample, which corresponds to about 18% of the average interest rate in the data, and the average marginal cost is about 2.40%.⁵³

I next use the estimated marginal costs as a dependent variable in equation (18) to identify the cost of capital regulation in the mortgage market, controlling for product characteristics and other supply-side factors. Table V shows the structural supply parameters. The main parameter of interest captures the effect of risk-weighted capital requirements on the marginal costs. I identify this effect by exploiting variation in risk-weighted capital requirements across lenders, across leverage levels, and over time. Column (1) of Table V shows the unconditional effect of capital regulation on the marginal cost. I find that a 1-percentage-point higher risk-weighted capital requirement increases the marginal cost of lending to first-time buyers by about 20 basis points.

In column (2) of Table V, I control for product characteristics and other supply-side factors that affect the cost of issuing mortgages. The effect of risk-weighted capital requirements on lender marginal cost of originating mortgages is still significant. A 1-percentage-point higher risk-weighted capital requirement increases the marginal cost of lending to first-time buyers by approximately 26 basis points, controlling for differences across lenders, which are common among products (lender shocks), and for differences across products, which are common across lenders (macroshocks).

In column (3) of Table V, I report results after including interacted lender-quarter fixed effects. In this specification, I only exploit the variation in risk-weighted capital requirements within a lender-time pair, which rules out concerns about other time-varying lender-specific factors affecting the cost of issuing mortgages. The coefficient that captures the impact of regulation on the cost of lending is still significant and the magnitude is slightly larger than, but not statistically different from, that in column (2).

In columns (4) and (5) of Table V, I explore the heterogeneity in the cost of risk-weighted capital requirements across lenders. In column (4), I

⁵³ Table IA.VII in the [Internet Appendix](#) breaks down markups and marginal costs by lender, LTV, and interest rate type. Product-level estimates of markups for the mortgage market are not available, but Button, Pezzini, and Rossiter (2010) decompose new lending rates for U.K. mortgages into funding costs, capital costs, and a residual. They find that after the financial crisis in 2008, the residual, which includes operating costs and markup, rose. As operating costs are unlikely to have changed, and if anything may have decreased as a result of consolidation, their finding is consistent with increasing markups.

Table V
Structural Supply Estimates

The table shows the structural parameters of the supply model from equation (18). The dependent variable is the marginal cost at the product level. Risk weights are the regulatory risk weights expressed in percentage terms. The capital requirements include both minimum requirements under Basel II (Pillar I, or 8% of risk-weighted assets) as well as lender-specific supervisory additions (Pillar II). High-LTV is a dummy equal to 1 for products with a maximum LTV above 85%. Fix 5 is a dummy for mortgages with a fixed period of five years. Swap rates are euro interest rate swaps for two and five years. Funding spread is a measure of the cost of lenders' wholesale funding based on secondary market senior unsecured bond spreads. High buffer (high funding spread) is a dummy for lenders with a capital buffer (wholesale funding spread) above the median. In column (6) I instrument risk-weighted capital requirements with a dummy equal to one after 2015 for lenders who experience an increase in capital requirements at the end of 2015. Robust standard errors are in parentheses. *, **, and *** indicate significance at the 0.10, 0.05, and 0.01 levels, respectively.

	Main			Heterogeneity		IV
	(1)	(2)	(3)	(4)	(5)	(6)
RW Capital Req (%)	0.209*** (0.041)	0.258*** (0.036)	0.297*** (0.045)	0.312*** (0.034)	0.233*** (0.019)	0.273** (0.136)
High LTV		1.038*** (0.083)	0.989*** (0.090)	1.034*** (0.039)	1.022*** (0.038)	0.904*** (0.248)
Fix 5		0.598*** (0.073)	0.592*** (0.072)	0.603*** (0.052)	0.598*** (0.052)	0.608*** (0.082)
Swap rates		0.280* (0.160)	0.297** (0.150)	0.278** (0.133)	0.281** (0.133)	0.219 (0.207)
Funding spreads		0.120 (0.210)		0.149 (0.194)	0.111 (0.191)	0.428 (0.281)
RW Capital Req (%) × High buffer				−0.083** (0.035)		
× High funding spread					0.143*** (0.031)	
Time F.e.	No	Yes	No	Yes	Yes	Yes
Lender F.e.	No	Yes	No	Yes	Yes	Yes
Time-Lender F.e.	No	No	Yes	No	No	No
F-stat						10.49
Marginal Cost (mean)	2.42	2.42	2.42	2.42	2.42	2.40
R ²	0.12	0.81	0.83	0.82	0.82	0.84
Observations	1,046	1,046	1,046	1,046	1,046	416

interact capital requirements with a dummy equal to 1 for lenders with a high capital buffer, defined as the difference between capital resources and capital requirements. In column (5), I interact capital requirements with a dummy equal to one for lenders with a high funding spread. I find that lenders with a higher buffer are less affected, while lenders with a higher funding spread are more affected, by the risk-weighted capital requirements. Both heterogeneity results are consistent with the literature on financial frictions, which suggests

that capital constraints matter more for banks that are less capitalized in bad times, as proxied in my setting by a low buffer and a high funding spread (Gambacorta and Mistrulli (2004), Kashyap, Stein, and Hanson (2010), Aiyar et al. (2014), Jiménez et al. (2017)).

Finally, in column (6) of Table V, I report estimates of equation (18) using the instrumental variable strategy. I estimate a difference-in-difference specification in which I instrument risk-weighted capital requirements with a dummy equal to 1 after 2015 for lenders who experience an increase in capital requirements at the end of 2015. The instrumental variable specification yields point estimates for the effect of risk-weighted capital requirements on lenders' marginal costs of around 27 basis points. The effect is slightly larger, but not significantly different from the baseline estimates in column (2) of Table V.

While the coefficient capturing the effect of leverage regulation on marginal costs is the main supply-side parameter of interest, the coefficients on product characteristics and other supply-side cost shifters provide an important check on the plausibility of the cost estimates. As expected, I find that higher leverage and longer fixed-rate mortgages have higher marginal costs. Even controlling for the duration of the fixed period, larger time-varying swap rates have a positive and marginally significant effect on the cost of mortgage lending. Finally, higher funding spreads increase lenders' costs of originating mortgages, but the estimates are noisy.

B.1. Discussion on Magnitude

The baseline estimate in column (2) of Table V shows an increase in the marginal cost of about 26 basis points for a 1-percentage-point increase in the risk-weighted capital requirement. Given an average marginal cost of 2.4%, marginal cost increases by approximately 11% for the average mortgage product.

To date there is no consensus on the impact of changes in capital requirements on the cost for lenders (Kovner and Van Tassel (2018)). Recent studies using different identification strategies and assumptions find estimates ranging from 3 basis points (Kisin and Manela (2016)), to 25 to 45 basis points (Hanson, Kashyap, and Stein (2011)), up until 60 to 90 basis points (Baker and Wurgler (2015)). To put my estimates of the cost of capital regulation into context, I simulate a 10-percentage-point increase in capital requirements, along the lines of previous studies (Hanson, Kashyap, and Stein (2011), Firestone, Lorenc, and Ranish (2017)).⁵⁴

I find that increasing capital requirements by 10 percentage points increases marginal costs by about 60 basis points in the mortgage market. This effect

⁵⁴ Table IA.VIII in the Internet Appendix shows the effects of a 10-percentage-point increase in capital requirements on marginal costs, rates, and competition. Note that given an average risk weight of 0.27, a 10-percentage-point increase in capital requirements corresponds to approximately a 3-percentage-point increase in risk-weighted capital requirements.

lies on the upper end of previous estimates and can be interpreted as an *upper bound* on the cost of a common increase in capital requirements. First, my estimates are not directly comparable because I do not impose a particular capital structure model or exploit a single observed regulatory change keeping all else equal. Different from most previous studies on the topic, I infer the *perceived* total cost of complying with the regulation only from the rates that lenders charge to mortgage borrowers after changes in risk-weighted capital requirements. Second, I allow lenders to adjust to the new regulation *only* along one margin: repricing mortgages. In reality, lenders can lower deposit rates, issue new equity, and retain earnings, especially if the change in regulation is spread over time. These additional margins of adjustment likely decrease the cost of increasing capital requirements (Elliott (2009)).

VI. Counterfactual Leverage Regulations

In this section, I use the estimated model to study alternative leverage regulations and their equilibrium impact on mortgage rates, market structure, and risk. First, Section VI.A compares two alternative counterfactuals that remove the regulatory-risk-weights gap between large and small lenders that I document Section II.A. Specifically, I simulate a counterfactual in which all lenders calculate risk-weighted capital requirements using the standardized approach and an alternative counterfactual in which small and medium lenders also adopt a representative IRB model.

Second, Section VI.B analyzes the interactions between risk-weighted capital requirements and regulations limiting household leverage via LTVs. I impose a maximum LTV limit by removing from households' choice set all mortgages with leverage above 90% both in a counterfactual market with homogeneous capital requirements, as was the case before the financial crisis, and in the current regime with risk-weighted capital requirements.

A. Equilibrium Effects of Risk Weights

The reduced-form evidence discussed in Section II suggests that risk weights affect pricing and specialization. Here, I quantify the equilibrium effect of alternative counterfactual policies that address the regulatory-risk-weights gap between large banks and other competitors. First, I illustrate how current differences in capital regulation affect mortgage rates over and above other factors through the lens of the model.

Consider two mortgages with a fixed amount of \$1, the same fixed period, the same maximum LTV, and the same expected default and refinancing risks (both equal to zero for simplicity). One mortgage is offered by a large lender using an internal model for the calculation of risk weights, while the other is offered by a small lender that employs the standardized regulatory approach. Under these simplifying assumptions, using lenders' first-order conditions for the constrained problem (equation (9)), the difference in prices between the

two lenders for product j in a given quarter t and market m is given by⁵⁵

$$r_{j,small} - r_{j,large} = \overbrace{\lambda(\underline{K}_{small}\rho_{j,small} - \underline{K}_{large}\rho_{j,large})}^{\text{Regulation}} - \overbrace{\left[\frac{1}{\sum_i \left(\frac{\partial s_{ij,small}}{\partial r_{j,small}} \frac{1}{s_{ij,small}} \right)} - \frac{1}{\sum_i \left(\frac{\partial s_{ij,large}}{\partial r_{j,large}} \frac{1}{s_{ij,large}} \right)} \right]}^{\text{Market power}}. \quad (20)$$

The higher risk weights for the small lender translate into higher rates, *ceteris paribus*. If the elasticity of demand for products offered by large lenders is lower, due, for example, to brand power, there is also an incumbent advantage, which affects the price gap between large lenders and smaller competitors.

Table IA.VIII in the [Internet Appendix](#) explores the effect of a common increase in capital requirements ($\underline{K}_{it}$) for all lenders. I find that after a common increase in capital requirements, large lenders using IRB models raise mortgage interest rates by about 50% less than small banks using the standardized regulatory approach. As a result of the differential impact, the largest lenders gain market share from their smaller competitors, increasing concentration in the mortgage market.

Here I explore the consequences of addressing the heterogeneity in regulations by changing how lenders compute risk weights (ρ_{jt}). Figure 8 illustrates two alternatives. First, I simulate an equilibrium without IRB models for the calculation of risk weights, which I label “All Standard.” Second, I allow lenders using a standardized approach to develop an internal model, using the average risk weights of large lenders. I label this second counterfactual case “All Internal.” Table VI reports the results for several variables of interest in a random subset of the first-time buyer market.⁵⁶

Panel A shows the effects of removing heterogeneity in risk-weighted capital requirements on market structure. I measure concentration in the market looking at both the Herfindahl index (HI) and the share of the six largest lenders for mortgage originations. The abolition of internal models leads the market to become more competitive. Large lenders lose the regulatory advantage (first element in equation (20)) and increase their prices following an increase in the risk weights for mortgages they originate. As a result of the higher rates, large lenders lose market share to smaller lenders already adopting the standardized regulatory approach. The share of the six largest lenders drops by more than 20% from almost 85% to less than 70%. Small lenders’ adoption of internal models also has a procompetitive effect on the market. I

⁵⁵ Large and small lenders’ marginal costs of issuing a mortgage may differ for reasons other than regulation. In the estimation, I control for differences across lenders and products in the marginal costs. In the counterfactual analyses, I keep these other factors constant.

⁵⁶ Table IA.IX in the [Internet Appendix](#) presents the results for the same variables computed separately for high- and low-LTV mortgages. Table IA.X in the [Internet Appendix](#) presents results of a robustness exercise in which I double the effect of rates on default.

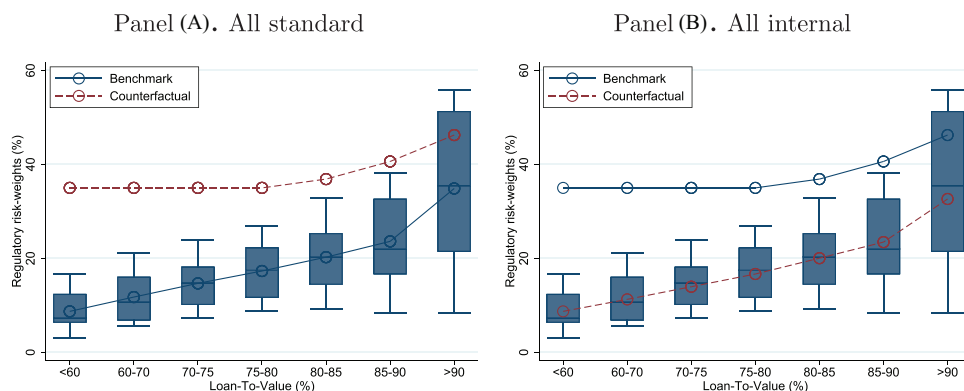


Figure 8. Counterfactual risk-weighted capital requirements. The figure shows the distribution of risk weights in the two counterfactual scenarios for capital requirements. In Panel (A), I show the first counterfactual, in which all lenders adopt the standard approach for setting risk weights. In Panel (B), I show the second counterfactual, in which I compute the mean risk-weight across lenders using the internal model and assign it to the small lenders using the standardized approach. (Color figure can be viewed at wileyonlinelibrary.com)

find that the HI declines from 16% to about 13%, as smaller lenders reduce prices and gain about 14 percentage points of market share, following the decrease in regulatory costs. The redistribution from large to small lenders is overall slightly less pronounced than in the first counterfactual.

Panel B of Table VI looks at aggregate costs, prices, and demand. I find that eliminating IRB models increases costs in the market by about 53 basis points, which are passed onto borrowers via higher initial rates. The latter increases by approximately 53 basis points, or about 20% relative to the baseline rate. As a result, the Lerner index decreases by about 16%, from a baseline of about 20 to about 17. As a result of higher mortgage prices, product demand decreases by 14%, as almost 800 borrowers switch to the outside option. The average loan size decreases by approximately £1,600, which is slightly higher than 1% of the average baseline loan amount.

In the all-IRB model counterfactual, marginal costs in the mortgage market decrease by about 15 basis points, as a result of lower capital requirements for small lenders. This decline translates into a reduction in prices of about 15 basis points. Overall, in this scenario, the Lerner index increases by about 8%. The slightly lower rates yield an increase in mortgage demand of slightly more than 1%, while there is almost no difference in loan amounts.

I use the model to compute a measure of consumer surplus based on the sum of indirect utilities and lenders' profits.⁵⁷ Panel C of Table VI reports the results. When all banks adopt the standardized approach, as a result of overall higher prices, average consumer surplus decreases by more than 30%; while

⁵⁷ In Section III of the Internet Appendix, I discuss the derivation of expected consumer surplus and relevant references to the literature. Lenders' profits are computed using equation (6).

Table VI
Counterfactual Risk-Weighted Capital Requirements

The table shows the baseline estimates of the model and two counterfactuals in a market for first-time buyers. In the first counterfactual scenario, all lenders adopt the standard approach for setting risk weights. In the second counterfactual, I compute the mean risk-weight across IRB lenders and simulate a scenario in which lenders using the standardized approach develop an IRB model that gives them the average risk-weight of their IRB competitors. Cost is the marginal cost in percentage points; price is the interest rate in percentage points; product demand is the total number of borrowers; consumer surplus is the log sum of the indirect utility of a representative consumer (see Section III of the [Internet Appendix](#)); profits is the average profit across lenders in thousand £; default is the average number of defaults in percentage points; and buffer is the difference between equity and the predicted loss. Others include challengers and building societies. Value is the actual value in the benchmark; Δ is the absolute change of the value in the counterfactual relative to the benchmark; and $\Delta(\%)$ is the percentage change of the value in the counterfactual relative to the benchmark.

	Baseline	Counterfactuals			
		All Standard		All Internal	
		Value	Δ	$\Delta(\%)$	Δ
Panel A: Market Structure					
Herfindahl index	16.01	−3.80	−23.7	−3.40	−21.2
Share top six	85.52	−18.11	−21.2	−13.98	−16.3
Panel B: Costs, Prices, and Demand					
Cost	2.14	0.53	24.5	−0.15	−6.9
Price	2.62	0.53	20.2	−0.15	−5.9
Lerner index	19.50	−3.15	−16.2	1.65	8.5
Product demand	5,638	−792	−14.0	86	1.5
Loan amount	135	−1.57	−1.2	0.52	0.4
Panel C: Borrowers and Lenders					
Consumer surplus	7,250	−2,578	−35.6	581	8.0
Lender profits	10,404	−1,507	−14.5	127	1.2
Panel D: Risk					
Default	1.34	0.11	8.5	−0.04	−3.0
Buffer:					
All	2.18	2.12	97.3	−0.09	−4.2
Top six	1.88	2.27	121.1	−0.01	−0.6
Others	4.02	0.57	14.2	−1.40	−34.9

when all banks adopt an internal model, the lower prices increase consumer surplus by about 8%. Quarterly lenders' profit decrease by approximately £1.5 million following the adoption of the standard approach, or almost 15% relative to the baseline. As a result of the adoption of IRB models by medium and small lenders, overall profit increases by about 1.2%.

Even if a full evaluation of the policy from a systemic point of view would require a general equilibrium approach, the model can be used to learn the effects of changing capital regulation on risk in the mortgage market and its differential impact on large systemic lenders. In Panel D of Table VI, I first look at borrower default. The expected default predicted by the model in the baseline case is about 1.3%, which is in line with the data (see Table I). With the abolition of IRB models, I observe an increase in the average default in the mortgage market by approximately 8%, as higher prices make it harder for borrowers to service their monthly payments. In the second counterfactual, lower prices and relatively small changes in borrower demand translate into a lower default rate, which decreases by approximately 3%.

Finally, in Panel D of Table VI, I also report a measure of resilience for the overall mortgage market. I compute the equity buffer as the difference in pounds between lenders' equity and expected losses for each mortgage. Lenders' equity is given by the endogenous loan size multiplied by the lender-specific capital requirement and the counterfactual risk weights; expected losses come from expected default given by equation (17) after lenders reoptimize rates in reaction to the new risk-weighted capital requirements and borrowers adjust demand to the new prices. In the baseline the equity buffer is about £2.2, with a large difference between the six top lenders and other lenders. The largest lenders have a buffer of about £1.9, which is less than half the buffer of other lenders. This difference is driven by large gaps in risk weights and small differences in observed ex-post performance, consistent with the motivating facts discussed in Section II.

Moving to the counterfactuals, I find that abolishing internal models doubles the equity buffer in the mortgage market, as large lenders experience an increase in their risk weights, especially for low-risk mortgages. In the all IRB model counterfactual, I find a small reduction in the extra buffer in the economy of about 4%, which is driven exclusively by small lenders, which experience a large drop in risk weights, especially for low-risk mortgages. However, the buffer of small lenders remains positive and higher than that of large lenders, which experience almost no change as a result of the policy. Given the central role played by large lenders in a crisis, my second counterfactual suggests that the reduction of small lenders' risk weights will not threaten the stability of the system (Acharya, Engle, and Richardson (2012), Akerlof et al. (2014), Bianchi (2016)). On the contrary, the procompetitive effect will decrease the intermediation conducted by large lenders, thus reducing their systemic importance.

To summarize, I find that heterogeneous capital regulation accounts for about 20% of the concentration in the market. The abolition of internal models addresses the imbalance between large and small lenders in terms of capital requirements, but the higher risk weights increase rates, reducing demand and consumer surplus. Extending a representative internal model to small lenders could also address the competitive distortion due to the differential regulatory

treatment, with limited impact on consumer surplus, lender profits, ex-post mortgage defaults, and the resilience of large lenders.⁵⁸

While the quantitative estimates cannot be applied outside the specific context I study, the insight that heterogeneity in leverage regulation affects competition in the mortgage market, with implications for borrower surplus, lender profits, and risk, goes beyond the U.K. mortgage market. In several countries in Continental Europe and in Australia where lenders' mortgage funding is deposit-based and there is no government insurance, it is likely that the forces I uncover for the U.K. mortgage market also hold.⁵⁹ For example, Behn, Haselmann, and Vig (2016b) find that in Germany, large banks using IRB models expanded lending to firms, potentially at the expense of smaller banks. In other countries such as the United States, the large recourse to securitization via GSEs and the emergence of shadow banks open up different margins of adjustment, which can impact the response of traditional banks to changes in risk-weighted capital regulation (Buchak et al. (2018a)).

B. Interactions between Borrower- and Lender-Based Leverage Regulations

An alternative way to regulate leverage in the mortgage market is to impose explicit limits on how much households can borrow. In this section, I use the estimated structural model to study the equilibrium effects of regulation that limits household leverage via LTV limits. Recent papers have explored similar policies, such as maximum DTI limits in the United States (DeFusco, Johnson, and Mondragon (2020)), and both LTI and LTV limits in Ireland (Acharya et al. (2020)). I contribute to this literature by providing a framework that allows one to study alternative policies under consideration. For example, the effects of imposing a maximum LTV limit at 80% relative to 90%. Specifically, I study the interaction of maximum LTV limits on borrowers and the capital requirement regime in place for lenders. In this way, I shed light on the equilibrium effects of jointly regulating both lenders' and borrowers' leverage.

Figure 9 shows the distribution of market shares by LTV in the baseline and after the elimination of mortgages with leverage above 90%. Panel (A) shows the equilibrium with a common capital requirement of 8% and a risk-weight of 50% for all lenders, as was the case before the global financial crisis during the Basel I regime, while Panel (B) shows the equilibrium with the actual risk-weighted capital requirements. As expected, mortgages with LTVs close

⁵⁸ The results in the second counterfactual are based on the assumption that the cost of developing the internal model is paid by the regulator. In practice, the implementation of this policy may involve, for example, the development of an internal model by the central bank using data provided by private lenders. If the burden falls on lenders, the cost of implementation may be too high as suggested by the observation that the majority of small and medium lenders do not adopt an IRB model. The potential benefits, however, can be large. A back-of-the-envelope calculation shows that the small lender that most gains from the model will see an increase in profits in the mortgage market equivalent to 22% of its annual profits.

⁵⁹ Among others, Lea (2010) and Seru (2019) provide detailed discussions of differences in regulations and institutional characteristics across mortgage markets around the globe.

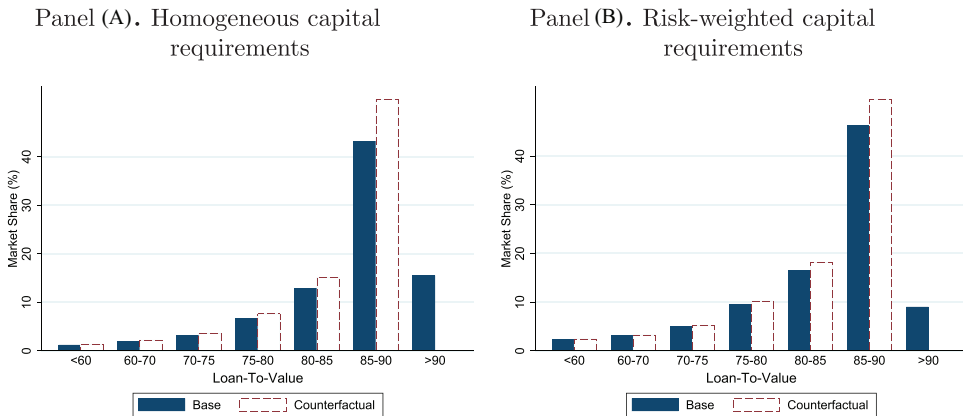


Figure 9. Counterfactual leverage limits. The figure shows the market shares of mortgages in the two counterfactual scenarios for leverage regulation. In both counterfactuals, I impose a maximum leverage limit of 90%, by excluding products with a maximum leverage above 90% from the choice set of all borrowers. In Panel (A), I show the counterfactual with homogeneous capital requirements, and in Panel (B), I show the equilibrium with risk-weighted capital requirements. (Color figure can be viewed at wileyonlinelibrary.com)

to 90% experience the largest increase, but as prices adjust, other products are affected in equilibrium. Table VII reports the quantification of costs and benefits and explores potential unintended consequences due to the interaction of multiple leverage regulations.

Panel A shows marginal costs, mortgage rates, and borrowers' demand. With homogeneous capital requirements at 8% and a 50% risk weight, both marginal costs and mortgage rates are higher than under the observed risk weighted capital requirements.⁶⁰ Mortgage rates are on average almost 3% with homogeneous capital regulations and about 2.6% in the current regime with risk-weighted requirements. As a result of eliminating high-leverage-high-cost mortgages, marginal costs go down and interest rates follow. The reduction in costs and interest rates is larger in the current regime with risk-weighted capital requirements, as high-leverage mortgages require more equity funding because of the higher risk weights. With homogeneous capital requirements, the decline in rates is about 4.5%, while with risk-weighted requirements, the decline is almost 8%. Despite the overall lower rates, there is a reduction in product demand, which drops between 1% and 4%.⁶¹

⁶⁰ Figure IA.14 in the Internet Appendix shows the average risk-weighted capital requirements across all lenders in the two scenarios with homogeneous and risk-weighted capital requirements, before and after the introduction of the leverage regulation on LTVs. While capital requirements are on average higher in the new regime, risk-weighted capital requirements for mortgages are lower.

⁶¹ This decrease can be seen as a lower bound on the true decrease in originations, as in the model, only a small fraction of borrowers (less than 5%) will be affected by this regulation. As I show in Table IA.VI of the Internet Appendix, my model slightly underpredicts mortgages with a maximum LTV above 90% with respect to the data.

Table VII
Counterfactual Leverage Limits

The table shows the baseline estimate of the model and two counterfactuals in a market for first-time buyers. In both counterfactuals, I impose a maximum LTV limit of 90%. Cost is the marginal cost in percentage points; price is the interest rate in percentage points; product demand is the total number of borrowers; consumer surplus is the log sum of the indirect utility of a representative consumer (see Section III of the [Internet Appendix](#)); profits is the average profit across lenders in thousand £; default is the average number of defaults in percentage points; and buffer is the difference between equity and the predicted loss. Value is the actual value in the benchmark; Δ is the change of the value in the counterfactual relative to the benchmark; and $\Delta(\%)$ is the percentage change of the value in the counterfactual relative to the benchmark.

	Homogenous Capital Regulation (Precrisis)			Risk-Weighted Capital Regulation (Postcrisis)		
	Baseline	No LTV > 90		Baseline	No LTV > 90	
	Value	Δ	$\Delta(\%)$	Value	Δ	$\Delta(\%)$
Panel A: Costs, Prices, and Demand						
Cost	2.50	-0.13	-5.16	2.14	-0.19	-8.89
Price	2.98	-0.13	-4.51	2.62	-0.20	-7.58
Product demand	5,195	-183	-3.53	5,638	-62	-1.09
Panel B: Borrowers and Lenders						
Consumer surplus	6,007	-750	-12.48	7,364	-390	-5.29
Lender profits	9,479	-349	-3.68	10,404	-171	-1.64
Panel C: Risk						
Default	1.39	-0.05	-3.58	1.34	-0.07	-5.27
Buffer						
All	3.96	0.00	0.04	2.18	-0.09	-4.35
Top six	3.95	0.00	0.05	2.00	-0.11	-5.38
Others	3.96	0.00	0.04	4.31	-0.05	-1.17

Panel B of Table VII shows the effect on borrowers and lenders. The reduction in consumer surplus as a result of the policy is large despite the decrease in prices, which should increase it. The large negative effect is driven not only by borrowers switching to the outside option, but also by the positive valuation that households attach to high-leverage mortgages, as I show in Table II. Consumer surplus declines by about 12% and 5% in the homogeneous and risk-weighted capital regime, respectively. Lender profits drop by around 3.7% when capital requirement is homogeneous, while the decrease is 2% under risk-weighted capital requirements.

Finally, Panel C of Table VII shows how regulating household leverage with LTV limits affects risk in the mortgage market. The limit to high-leverage mortgages decreases defaults, which drop by about 3% to 5% in both cases as a result of lower prices and lower leverage. To capture the overall risk in

the mortgage market, I also look at lenders' equity buffers. In the counterfactual with homogeneous capital requirements, there is almost no change in the buffer or a slightly positive change, as lower prices reduce defaults. In the scenario with risk-weighted capital requirements, there is a 4% decrease in the equity buffer in the market as a result of the ban on mortgages with the highest risk but also the highest risk weights. According to my estimates, the decline in the buffer is even larger for the largest lenders, which experience a drop of more than 5%, thus increasing their exposure to risk in the mortgage market.

To summarize, I find that a regulation targeting LTV can be effective in reducing borrower defaults, but with large impacts for borrowers.⁶² Furthermore, I show that the interaction in equilibrium between regulations on borrowers and lenders can have distributional effects and unintended consequences. The larger effects of abolishing high-LTV mortgages on consumer surplus and lender profits under homogeneous capital requirements are due to the fact that low-LTV borrowers are "subsidizing" high-LTV borrowers, while the opposite occurs under risk-weighted capital requirements, given the observed risk weights and defaults. Finally, leverage limits applied in a market with risk-weighted capital regulations can reduce the equity buffer of large lenders, thereby increasing systemic risk. This result emphasizes the importance of looking at equilibrium effects, especially when multiple regulations are in place and the implication of changing one may depend on other policies.

VII. Conclusion

Leverage regulation has been at the center of the academic and policy debate since the global financial crisis, and there is an ongoing effort to better understand the channels through which it operates and to evaluate its effectiveness. In this paper, I focus on leverage regulation in the U.K. mortgage market, in which lenders face different capital regulations and limits to household leverage have been recently introduced. I develop a tractable equilibrium model of the mortgage market that accounts for several features characterizing borrower demand and lender competition, and I estimate the model using a new identification strategy that exploits variation in risk-weighted capital requirements both across lenders and within lender across mortgages with different LTVs.

I quantify the cost of capital regulation for lenders and find that a 1-percentage-point increase in risk-weighted capital requirements raises lenders' costs by 26 basis points or 11% for the average mortgage product when lenders can react only by adjusting their price. With the estimated model, I show that heterogeneous leverage regulation shapes market structure. In particular, removing the policy-driven difference in risk weights can promote competition without exacerbating risk. Finally, I find that LTV limits reduce

⁶² This finding echos the results in DeFusco, Johnson, and Mondragon (2020), who find that the introduction of a downpayment-to-income limit in the United States led to a large reduction in demand and a more limited effect on defaults.

borrower defaults but have a large impact on originations and consumer surplus, and can potentially reduce large lenders' risk-weighted equity buffers, and thereby affect systemic risk.

My paper can be extended in several directions. The lenders' problem can be expanded to account for the acceptance/rejection margin. This would allow leverage regulation to affect loan supply not only through changes in loan rates, but also through changes in underwriting standards. I capture this channel in a reduced-form way through affordability constraints, as well as unobservable product characteristics. A more comprehensive model and empirical strategy that feature both pricing and rejection choices would be an interesting avenue for future research. Moreover, in this paper, I focus mostly on the costs of risk-weighted capital requirements, their transmission to interest rates, and their implications for market structure. It would be interesting to enrich the framework to account explicitly for endogenous default choices on both the demand and the supply side. Adding the default option for borrowers would allow for a comprehensive measure of consumer surplus, while lenders' bankruptcy choices would provide an explicit microfoundation for leverage regulation and a fully fledged quantification of the trade-offs.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1: Internet Appendix.
Replication Code.